Peering into the twilight - a systematic review on visual-based investigation of worldwide mesophotic ecosystems

Claudia Campanini

2026-03-31

Table of Contents

DATA PREPARATION 2
Load libraries 2
Load data 3
Cleaning 3
MISSING / ND / NOTAPP VALUES 3
GEOGRAPHICAL DISTRIBUTION 5
Provinces 5
Ecoregions 6
Realms 8
Latitudinal region 8
Export geographical distribution summary tables 9
COUNT COLUMNS WITH COMBO 10
Export summary tables 11
COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED) 14
COUNT COLUMNS SPLIT 17
COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED) 22
PUBLICATIONS PER DEPTH 26
Publications per depth 26
Publication per depth by latitudinalRegion 27
PLATFORM BY DEPTH AND latitudinalRegion 29
DIRECT - INDIRECT BY DECADE 36
FIELD & INDICATORS 42
Calculate # of ecological indicators per record 47

SAMPLING UNIT 48
Prepare a data frame for sampling unit 48
Calculate # of sampling unit per record 49
Simplified sampling units 50
Quadrat extracted from transect 54
DATA TYPE 54
HABIT 57
SUBSTRATE 58
BENTHIC POSITION 61
TARGET 64
Target level 64
Detected phyla in community studies 66
Target taxonomic groups not at the community level 70
New species described per phyla 73
New species descriptions that included molecular analyses 76
DATA CROSSING 77
VISUALLY SURVEYED SURFACE 77
CCR and latitudinalRegion 77
Observation 78
Temp replicates & exp 78
Health 78
ROV & sample 81
Molecular techniques in taxonomic and non-taxonomic studies 81
Most common target species 81

DATA PREPARATION

Load libraries

```
library(here)

## here() starts at C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analys
is/Projects/Mesophotic_systematic_lit_rev_analysis

library(dplyr)
```

```
##
## Caricamento pacchetto: 'dplyr'

## I seguenti oggetti sono mascherati da 'package:stats':
##
##     filter, lag

## I seguenti oggetti sono mascherati da 'package:base':
##
##     intersect, setdiff, setequal, union

library(ggplot2)
library(readxl)   # to import excel data sets
library(magrittr)
library(stringr) #many data wrangling functions
library(readxl)
```

Load data

```
raw <- read_excel(here("Extracted_data_analysis","Input_data","extracted_raw.
xlsx"))
```

Cleaning

Clean year columns

```
#keep the year only in the year column (in some rows exact time is specified)
raw$publicationYear <-  stringr::str_remove(raw$publicationYear,
                                  pattern = "\\-.*")
```

MISSING / ND / NOTAPP VALUES

Create a summary table

```
library(purrr)

##
## Caricamento pacchetto: 'purrr'

## Il seguente oggetto è mascherato da 'package:magrittr':
##
##     set_names

library(tibble)

# Get all column names
cols <- names(raw[,-c(1:13)])

# Create a summary list
summary_list <- map(cols, function(col) {
  data <- raw[,-c(1:13)][[col]]
```

```r
  tibble(
    Column     = col,
    unique     = paste(unique(data), collapse = ", "),
    nd         = sum(data == "nd", na.rm = TRUE),
    notapp     = sum(data == "notapp", na.rm = TRUE),
    assigned   = sum(!is.na(data) & data != "nd" & data != "notapp"),
    `NA`         = sum(is.na(data)),
    total      = sum(data == "nd", na.rm = TRUE) +
                 sum(data == "notapp", na.rm = TRUE) +
                 sum(!is.na(data) & data != "nd" & data != "notapp")
  )
})

# Combine into one dataframe
summary_df <- bind_rows(summary_list)

summary_df
```

```
## # A tibble: 72 × 7
##    Column                 unique                  nd notapp assigned  `NA`
 total
##    <chr>                  <chr>                <int>  <int>    <int> <int>
 <int>
##  1 meso_direct_indirect   indirect, direct, m…     5      0      989     0
   994
##  2 meso_samplingPlatform  acou+bruv+tcs, bruv…     5      0      989     0
   994
##  3 taxonomic_publication  NA, taxa, taxa++         0      0       75   919
    75
##  4 meso_substrate         hard, mixed, artifi…    32      0      889    73
   921
##  5 meso_habitat           coralreef, mixed, c…    21      0      918    55
   939
##  6 meso_protection        mixed, nd, yes, no     652      0      342     0
   994
##  7 taxonomicTargetLevel   comm, group, comm+g…     0     19      975     0
   994
##  8 habit                  mixed, sessile, vag…     0     17      977     0
   994
##  9 environmentalPosition  dem+epi, endo+epi, …     0      8      986     0
   994
## 10 detected_meso_seagrass NA, x, notapp            0     11       32   951
    43
## # ℹ 62 more rows
```

```r
rm(summary_list, cols)
```

GEOGRAPHICAL DISTRIBUTION

```
location <- raw[,c (
  "recordTitle","DOI","publicationYear","prov_code","ecoreg_code")]
```

Load Marine Ecoregion of the World data

```
meow <- read.csv(file=here("Extracted_data_analysis","Input_data","MEOW.csv"))
```

Provinces

```
prov <- location

prov_lvls <-
  unique(
    unlist(
      str_split(location$prov_code, "\\_")
                                )
                          )



prov[,prov_lvls] <- sapply(prov_lvls,
                                        function(y){
                                          str_count(location$prov_code, y)
                                          }
                                        )
rm(prov_lvls)
```

Create a column with the sum of provinces per publication

```
prov %<>%
   mutate(
     prov_per_record = rowSums(
     across(
       where(is.numeric)
       )
     )
   )

prov$prov_per_record <- as.factor(prov$prov_per_record)

prov_per_record_df <- prov %>%
  group_by(prov_per_record) %>%
  summarise(n_records = length(prov_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,0))

prov_per_record_df

## # A tibble: 10 × 3
##    prov_per_record n_records percentage
##    <fct>               <int>      <dbl>
```

```
##  1 1                       442          44
##  2 2                       167          17
##  3 3                       347          35
##  4 4                         7           1
##  5 5                        12           1
##  6 6                        15           2
##  7 7                         1           0
##  8 8                         1           0
##  9 11                        1           0
## 10 26                        1           0
```

Create a data frame with number of publication per province

```
prov_tot <-
  prov [, -(length(prov)-1)] %>% # filter out the sum column
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

prov_tot <- reshape2::melt(prov_tot, id.vars = NULL)

colnames(prov_tot) <- c("Province", "Count")

prov_tot %<>%
      mutate_at("Province",
                stringr::str_replace, "_1", "")
```

Ecoregions

```
ecoreg <- location
ecoreg_lvls <-
  unique(
    unlist(
      str_split(ecoreg$ecoreg_code, "\\_")
                                )
                          )

ecoreg_lvls<- ecoreg_lvls[! ecoreg_lvls %in% c("")]

ecoreg[,ecoreg_lvls] <- sapply(ecoreg_lvls,
                                        function(y){
                                          str_count(ecoreg$ecoreg_code, y)
                                          }
                                        )
```

Create a column with the number of ecoregions per study

```
ecoreg %<>%
   mutate(
```

```
    ecoreg_per_record = rowSums(
    across(
      where(is.numeric)
      )
    )
  )

ecoreg$ecoreg_per_record <- as.factor(ecoreg$ecoreg_per_record)
```

Calculate number of records by number of ecoregions in %

```
ecoreg_per_record_df <- ecoreg %>%
  group_by(ecoreg_per_record) %>%
  summarise(n_records = length(ecoreg_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

ecoreg_per_record_df
```

```
## # A tibble: 6 × 3
##   ecoreg_per_record n_records percentage
##   <fct>                 <int>      <dbl>
## 1 1                       877       88.2
## 2 2                        74        7.4
## 3 3                        30        3
## 4 4                         9        0.9
## 5 5                         3        0.3
## 6 12                        1        0.1
```

Create a data frame with number of publication per ecoregion

```
ecoreg_tot <-
  ecoreg [, -(length(ecoreg)-1)] %>% # filter out the sum column
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

ecoreg_tot <- reshape2::melt(ecoreg_tot, id.vars = NULL)

colnames(ecoreg_tot) <- c("Ecoregion", "Count")

ecoreg_tot %<>%
      mutate_at("Ecoregion",
                stringr::str_replace, "_1", "")
```

Add count column to Meow and set 0 to all the ecoregions where no study was conducted

```
colnames(ecoreg_tot)[colnames(ecoreg_tot) == 'Ecoregion'] <- 'ECO_CODE'
ecoreg_tot$ECO_CODE <- as.character(ecoreg_tot$ECO_CODE)
meow$ECO_CODE <- as.character(meow$ECO_CODE)
```

```
ecoreg_tot <-
  left_join(meow, ecoreg_tot, by = 'ECO_CODE') %>%
  mutate(Count = replace(Count, is.na(Count), 0))
```

Export count per ecoregion to csv for the map in QGIS

```
writexl::write_xlsx(ecoreg_tot,
           here("Extracted_data_analysis","global","Output-Transformed_data",
"eco_reg_tot.xlsx"))
```

Realms

```
realms <- ecoreg_tot %>%
    group_by(RLM_CODE, REALM) %>%
  summarise(Count=sum(Count))

## `summarise()` has regrouped the output.
## ℹ Summaries were computed grouped by RLM_CODE and REALM.
## ℹ Output is grouped by RLM_CODE.
## ℹ Use `summarise(.groups = "drop_last")` to silence this message.
## ℹ Use `summarise(.by = c(RLM_CODE, REALM))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

realms %<>%
  rename(Code = RLM_CODE,
         Realm = REALM)

writexl::write_xlsx(realms,
           here("Extracted_data_analysis","global","Output-Transformed_data",
"Realm_count.xlsx"))
```

Latitudinal region

Calculate the number of studies per latitude zone

```
env <- raw[,c (
  "File","latitudinalRegion")]

unique(env$latitudinalRegion)

## [1] "tropical"           "temperate"          "polar"
## [4] "temperate+tropical" "polar+temperate"

env_lvls <-
  unique(
    unlist(
      str_split(env$latitudinalRegion, "\\_")
                                )
                          )
```

```
env$latitudinalRegion[env$latitudinalRegion=="tropical+temperate"] <-"tempera
te+tropical"


env$latitudinalRegion <- as.factor(env$latitudinalRegion)

env[,env_lvls] <- sapply(env_lvls,
                                      function(y){
                                        str_count(env$latitudinalRegion, y)
                                        }
                                      )

env %<>% tidyr::replace_na(list(latitudinalRegion = "temperate"))

str(env)
```

```
## tibble [994 × 7] (S3: tbl_df/tbl/data.frame)
##  $ File              : chr [1:994] "AbdulWahabEtAl_2018_Biodiversity_spati
al_patterns.md" "AbesamisEtAl_2018_Benthic_habitat_fish.md" "AchilleosEtAl_20
20_Bryozoan_diversity_Cyprus.md" "AdamsEtAl_2020_Rhodolith_bed_discovered.md"
 ...
##  $ latitudinalRegion : Factor w/ 5 levels "polar","polar+temperate",..: 5
5 3 3 3 3 3 5 5 3 ...
##  $ tropical          : int [1:994] 1 1 0 0 0 0 0 1 1 0 ...
##  $ temperate         : int [1:994] 0 0 1 1 1 1 1 0 0 1 ...
##  $ polar             : int [1:994] 0 0 0 0 0 0 0 0 0 0 ...
##  $ temperate+tropical: int [1:994] 0 0 0 0 0 0 0 0 0 0 ...
##  $ polar+temperate   : int [1:994] 0 0 0 0 0 0 0 0 0 0 ...
```

```
env_tot<- as.data.frame(summary(env$latitudinalRegion))

env_tot %<>%
  mutate(Latitude_zone = as.factor(rownames(env_tot))) %>%
  rename(Count = "summary(env$latitudinalRegion)")


env_tot
```

```
##                     Count      Latitude_zone
## polar                  28              polar
## polar+temperate         1    polar+temperate
## temperate             523          temperate
## temperate+tropical     15 temperate+tropical
## tropical              427           tropical
```

Export geographical distribution summary tables

```
writexl::write_xlsx(list(env_tot,realms, prov_tot, prov_per_record_df, ecoreg
_tot, ecoreg_per_record_df),
```

```
          here("Extracted_data_analysis","global","Output-Transformed_data",
"geographical_distribution_summary.xlsx"))
```

COUNT COLUMNS WITH COMBO

Create a subset of columns

```
count_columns <- raw %>%
  select(File,
         meso_substrate,
         habit,
         environmentalPosition,
         observationalORexperimental,
         taxonomicTargetLevel,
         indicators,
         collectedDataType,
         molecularAnalyses)
```

Prepare data set with columns that need count only and add percentage column

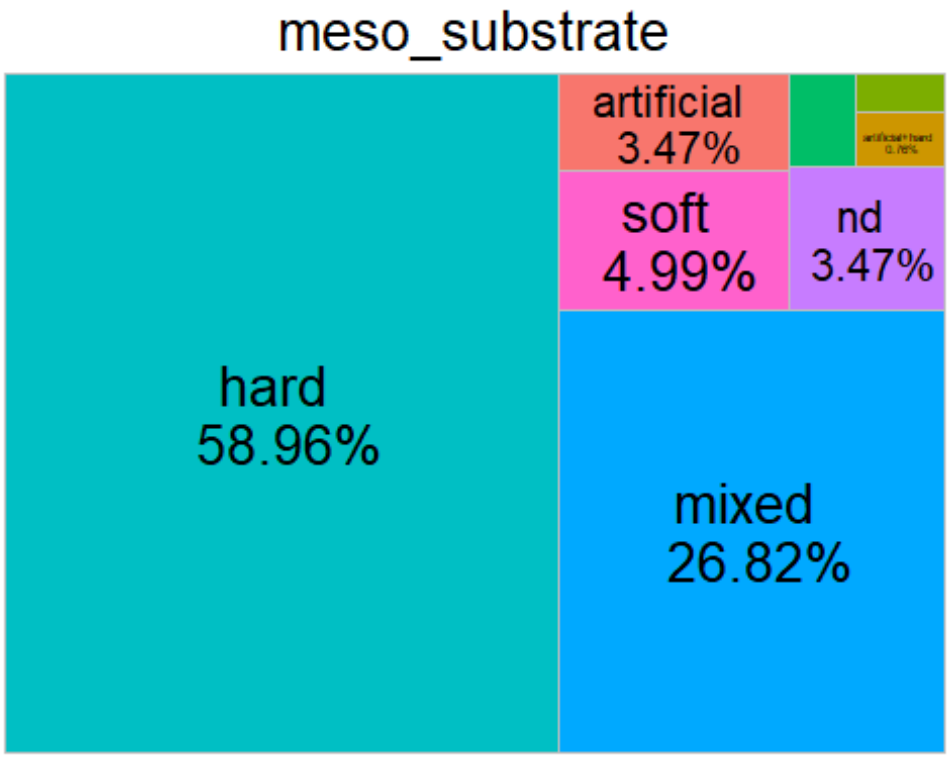
```
count_columns[,] <- lapply(count_columns[,], as.factor)

count_results <- list()

for (col_name in colnames(count_columns)) {
  counts <- count(count_columns, !!sym(col_name))

  count_results[[col_name]] <- counts

  colnames(count_results[[col_name]]) <- c(col_name, "Count")

  count_results[[col_name]] %<>%
    filter(
      if_all(.cols = everything(), ~ !grepl("notapp", .x))
    ) %>%
 # delete rows with "notapp"
    tidyr::drop_na()

  count_results[[col_name]] %<>%
    mutate(Percentage= round(Count / sum(Count) * 100,2))
}

rm(col_name, counts)

count_results <- count_results[names(count_results) != "File"]
```

Export summary tables

```
library(openxlsx)

write.xlsx(count_results,
           file = here("Extracted_data_analysis",
                       "global",
                       "Output-Transformed_data",
                     "summary_counts_question_1-3.xlsx"))
```

Create treemaps charts

```
treemap_func = function(dat,var) {
  ggplot(dat,
       aes(area=Count,
           fill=.data[[var]],
           line = "black",
           label=paste0(.data[[var]],
                        " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                     size = 20)+
  labs(title = var)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")

}

treemap_counts <- list()

for (col_name in colnames(count_columns[,-1])) {

  treemap_counts[[col_name]] <- treemap_func(dat= count_results[[col_name]],
var = col_name)

}

treemap_counts[[1]]
```



Save all tree maps plots

```
lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0(x,".tiff"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis",
                     "global",
                     "Output-Visualization",
                   "Treemap_counts"),
         create.dir = T
         )
  })
```

```
## [[1]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/meso_substrate.tiff"
##
## [[2]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/habit.tiff"
##
## [[3]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
```

```
Visualization/Treemap_counts/environmentalPosition.tiff"
##
## [[4]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/observationalORexperimental.tiff"
##
## [[5]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/taxonomicTargetLevel.tiff"
##
## [[6]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/indicators.tiff"
##
## [[7]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/collectedDataType.tiff"
##
## [[8]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/molecularAnalyses.tiff"
```

```r
lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0(x,".png"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",
                     "global",
                     "Output-Visualization",
                   "Treemap_counts"),
         create.dir = T
         )
  })
```

```
## [[1]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/meso_substrate.png"
##
## [[2]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/habit.png"
##
## [[3]]
```

```
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/environmentalPosition.png"
##
## [[4]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/observationalORexperimental.png"
##
## [[5]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/taxonomicTargetLevel.png"
##
## [[6]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/indicators.png"
##
## [[7]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/collectedDataType.png"
##
## [[8]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Treemap_counts/molecularAnalyses.png"
```

COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)

Remove purely taxonomic studies and subset columns (taxa++ was assigned to studies that led to the description of new species but where also ecological e.g. describing community for a given taxonomic group)

```
methods <- raw %>%
  filter(taxonomic_publication == "taxa++" |  is.na(taxonomic_publication) )
 %>%
  select(File,
         meso_design,
         meso_replicates,
         temporalReplicates,
         abioticVariablesCollection,
         abioticVariables)

count_columns_method <- methods
```

Prepare data set with columns that need count only and add percentage column

```
count_columns_method[,] <- lapply(count_columns_method[,], as.factor)

count_results_method <- list()

for (col_name in colnames(count_columns_method)) {
  counts <- count(count_columns_method, !!sym(col_name))

  count_results_method[[col_name]] <- counts

  colnames(count_results_method[[col_name]]) <- c(col_name, "Count")

  count_results_method[[col_name]] %<>%
    filter(
      if_all(.cols = everything(), ~ !grepl("notapp", .x))
    ) %>% # delete rows with "notapp"
    tidyr::drop_na()

  count_results_method[[col_name]] %<>%
    mutate(Percentage= round(Count / sum(Count) * 100,1))
}

rm(col_name, counts)

count_results_method <- count_results_method[names(count_results_method) != "
File"]
```

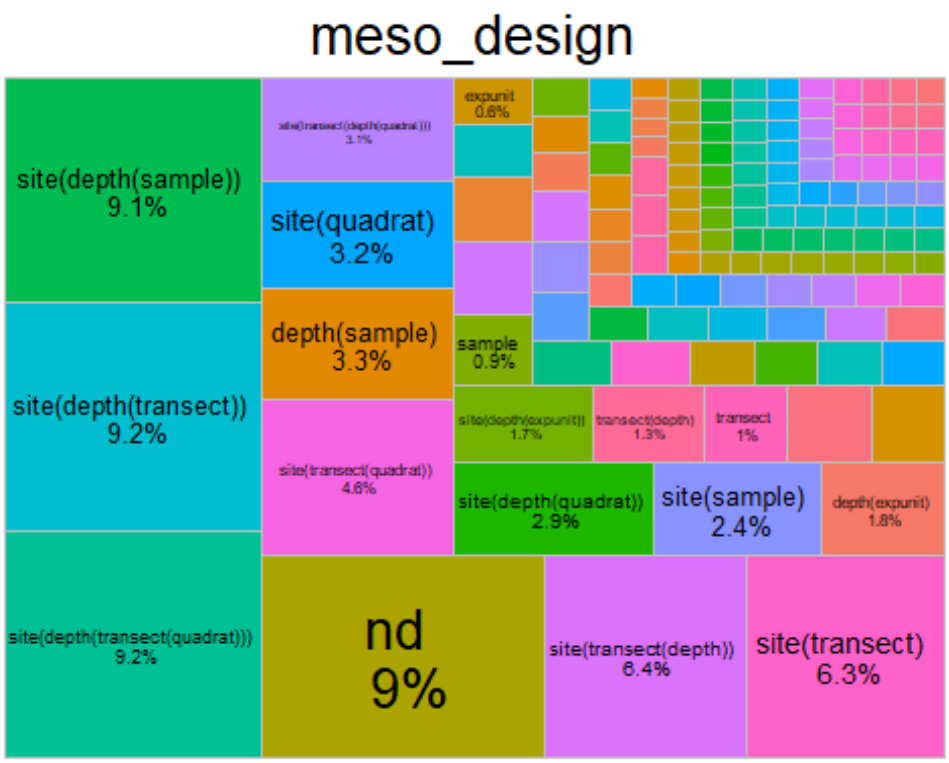
Export summary tables

```
library(openxlsx)

write.xlsx(count_results_method,
           file = here("Extracted_data_analysis",                        "glo
bal",                          "Output-Transformed_data",
                     "summary_counts_question_4.xlsx"))
```

Create treemaps charts

```
treemap_func = function(dat,var) {
  ggplot(dat,
       aes(area=Count,
           fill=.data[[var]],
           line = "black",
           label=paste0(.data[[var]],
                        " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                     size = 20)+
```

```
  labs(title = var)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")

}

treemap_counts <- list()

for (col_name in colnames(count_columns_method[,-1])) {

  treemap_counts[[col_name]] <- treemap_func(dat= count_results_method[[col_n
ame]], var = col_name)

}

treemap_counts[[1]]
```



Save all tree maps plots

```
lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0("treemap",x,".tiff"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis","global",
   "Output-Visualization","Treemaps_methods"),
         create.dir = T
```

```
        )
  })

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0("treemap",x,".png"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",                        "globa
l",                        "Output-Visualization","Treemaps_methods"),
         create.dir = T
         )
  })
```

COUNT COLUMNS SPLIT

Create a list of data frame splitting values

```
split_columns <- raw %>%
  select(meso_substrate,
         habit,
         environmentalPosition,
         observationalORexperimental,
         taxonomicTargetLevel,
         indicators,
         collectedDataType,
         molecularAnalyses)


split_columns[,] <- lapply(split_columns[,], as.factor)
```

Prepare data set with columns

```
split_results <- list()

for (col_name in colnames(split_columns)) {
  # Split the values of each column by "+"
  lvls <- unique(unlist(str_split(split_columns[[col_name]], "\\+")))

  # Initialize a list to store counts for each level
  split_counts <- sapply(lvls, function(y) {
    # Count how many times each unique level appears in the column
    str_count(split_columns[[col_name]], fixed(y))
    }
    )

  split_results[[col_name]] <- split_counts
}
```

```
## Warning in stri_count_fixed(string, pattern, opts_fixed = opts(pattern)):
empty
## search patterns are not supported

rm(lvls)
```

Add percentage column

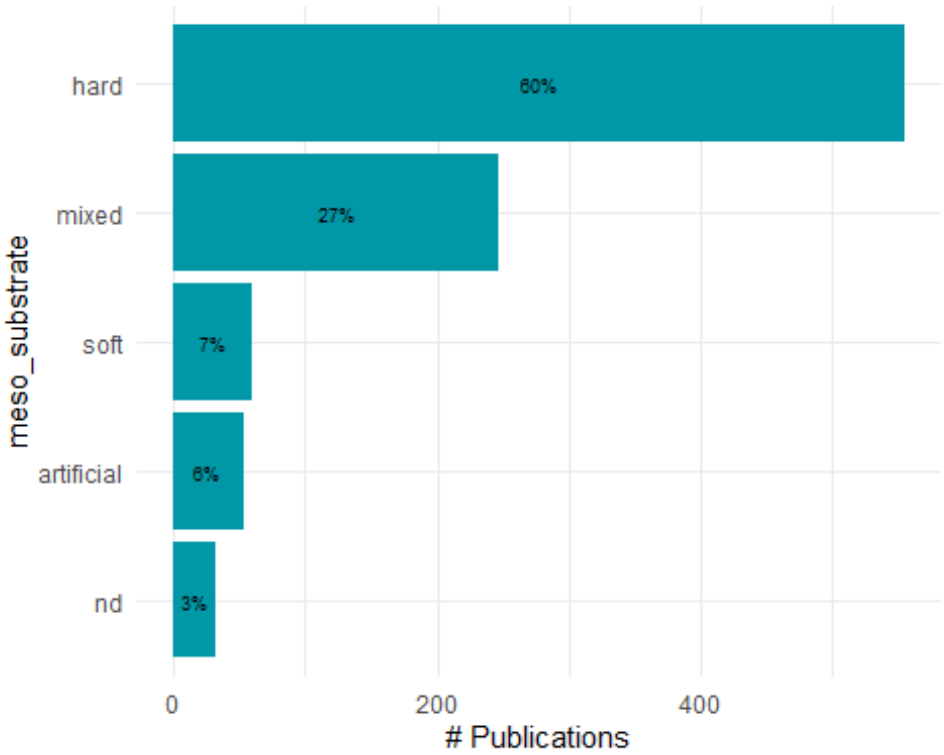
```
split_tot <- list()

for(col_name in colnames(split_columns)) {
  split_results[[col_name]] <- as.data.frame(split_results[[col_name]])

  split_results[[col_name]] %<>% # delete rows with missing values
    select(where(function(x) any(!is.na(x)))) %>%
    select(-any_of("notapp")) %>%
    tidyr::drop_na()

    split_tot[[col_name]] <- as.data.frame(split_results[[col_name]]) %>%
    summarise(across(where(is.numeric), list(sum)))

  split_tot[[col_name]]  <- reshape2::melt(split_tot[[col_name]], id.vars = N
ULL)

  colnames(split_tot[[col_name]]) <- c(col_name, "Count")

  split_tot[[col_name]] %<>%
        mutate_at(col_name,
                  stringr::str_replace, "_1", "")


  split_tot[[col_name]] %<>%
      mutate(Percentage= round(Count / nrow(split_results[[col_name]] ) * 100,
2))

}

rm(col_name)
```

Export summary tables

```
library(openxlsx)

write.xlsx(split_tot,
           file = here("Extracted_data_analysis",                        "glo
bal",                        "Output-Transformed_data",
                     "summary_split_counts_question_1_3.xlsx"))
```

Create bar charts for each collected parameter

```
barchart_func = function(dat,var) {

 ggplot(dat,
       aes(x = reorder(.data[[var]], Count), y = Count, fill= .data[[var]]))
+
  geom_bar(stat = "identity", fill = "#0097A7") +
  geom_text(
    data = dat %>% filter(Percentage >= 2),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust =0.5),
    color = "black",
    size = 2.5
  )+
  labs(y = "# Publications", x = var) +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")

}

barchart_counts <- list()

for (col_name in colnames(split_columns)) {

  barchart_counts[[col_name]] <- barchart_func(dat= split_tot[[col_name]], va
r = as.character(col_name))

}

barchart_counts[[1]]
```




```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".tiff"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis",                        "globa
l",                        "Output-Visualization","Barcharts_split"),
         create.dir = T
         )
  })
```

```
## [[1]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_meso_substrate.tiff"
##
## [[2]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_habit.tiff"
##
## [[3]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_environmentalPosition.tiff"
##
## [[4]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
```

```
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_observationalORexperimental.tiff"
##
## [[5]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_taxonomicTargetLevel.tiff"
##
## [[6]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_indicators.tiff"
##
## [[7]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_collectedDataType.tiff"
##
## [[8]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_molecularAnalyses.tiff"
```

```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".png"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",                        "globa
l",                        "Output-Visualization","Barcharts_split"),
         create.dir = T
         )
  })
```

```
## [[1]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_meso_substrate.png"
##
## [[2]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_habit.png"
##
## [[3]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_environmentalPosition.png"
##
## [[4]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
```

```
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_observationalORexperimental.png"
##
## [[5]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_taxonomicTargetLevel.png"
##
## [[6]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_indicators.png"
##
## [[7]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_collectedDataType.png"
##
## [[8]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/global/Output-
Visualization/Barcharts_split/bar_molecularAnalyses.png"
```

COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)

Create data frame with subset

```
split_columns_methods <- methods

split_columns_methods[,] <- lapply(split_columns_methods[,], as.factor)
```

Prepare data set with columns

```
split_results_methods <- list()

for (col_name in colnames(split_columns_methods)) {
  # Split the values of each column by "+"
  lvls <- unique(unlist(str_split(split_columns_methods[[col_name]], "\\+")))

  # Initialize a list to store counts for each level
  split_counts_methods <- sapply(lvls, function(y) {
    # Count how many times each unique level appears in the column
    str_count(split_columns_methods[[col_name]], fixed(y))
    }
    )

  split_results_methods[[col_name]] <- split_counts_methods
```

```
}

rm(lvls)
```

Add percentage column

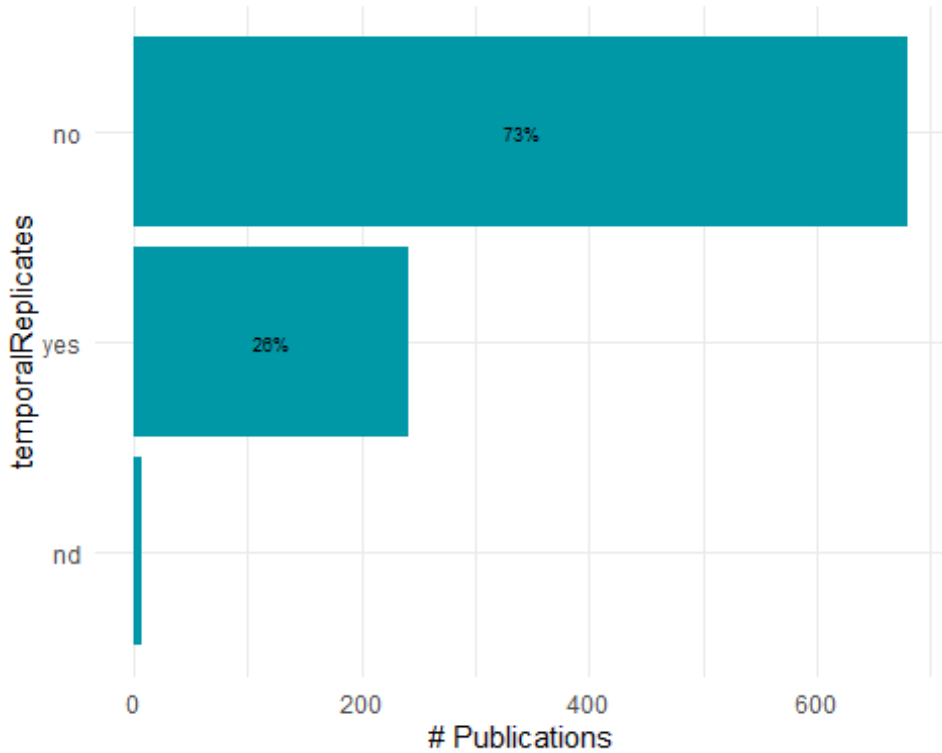
```
split_tot_methods <- list()

for(col_name in colnames(split_columns_methods)) {
  split_results_methods[[col_name]] <- as.data.frame(split_results_methods[[c
ol_name]])

  split_results_methods[[col_name]] %<>% # delete rows with missing values
    select(where(function(x) any(!is.na(x)))) %>%
    select(-any_of("notapp")) %>%
    tidyr::drop_na()

    split_tot_methods[[col_name]] <- as.data.frame(split_results_methods[[col
_name]]) %>%
    summarise(across(where(is.numeric), list(sum)))

  split_tot_methods[[col_name]]  <- reshape2::melt(split_tot_methods[[col_nam
e]], id.vars = NULL)

  colnames(split_tot_methods[[col_name]]) <- c(col_name, "Count")

  split_tot_methods[[col_name]] %<>%
        mutate_at(col_name,
                  stringr::str_replace, "_1", "")


  split_tot_methods[[col_name]] %<>%
      mutate(Percentage= round(Count / nrow(split_results_methods[[col_name]]
 ) * 100,2))

}

rm(col_name)
```

Export summary tables

```
library(openxlsx)

write.xlsx(split_results_methods,
           file = here("Extracted_data_analysis",                        "glo
bal",                        "Output-Transformed_data",
                     "summary_split_counts_question_4.xlsx"))
```

Create bar charts per each parameter collected

```
barchart_func = function(dat,var) {

 ggplot(dat,
       aes(x = reorder(.data[[var]], Count), y = Count, fill= .data[[var]]))
+
  geom_bar(stat = "identity", fill = "#0097A7") +
  geom_text(
    data = dat %>% filter(Percentage >= 2),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust =0.5),
    color = "black",
    size = 2.5
  )+
  labs(y = "# Publications", x = var) +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")

}

barchart_split_methods <- list()

for (col_name in colnames(split_columns_methods)) {

  barchart_split_methods[[col_name]] <- barchart_func(dat= split_tot_methods
[[col_name]], var = as.character(col_name))

}

barchart_split_methods[[4]]
```



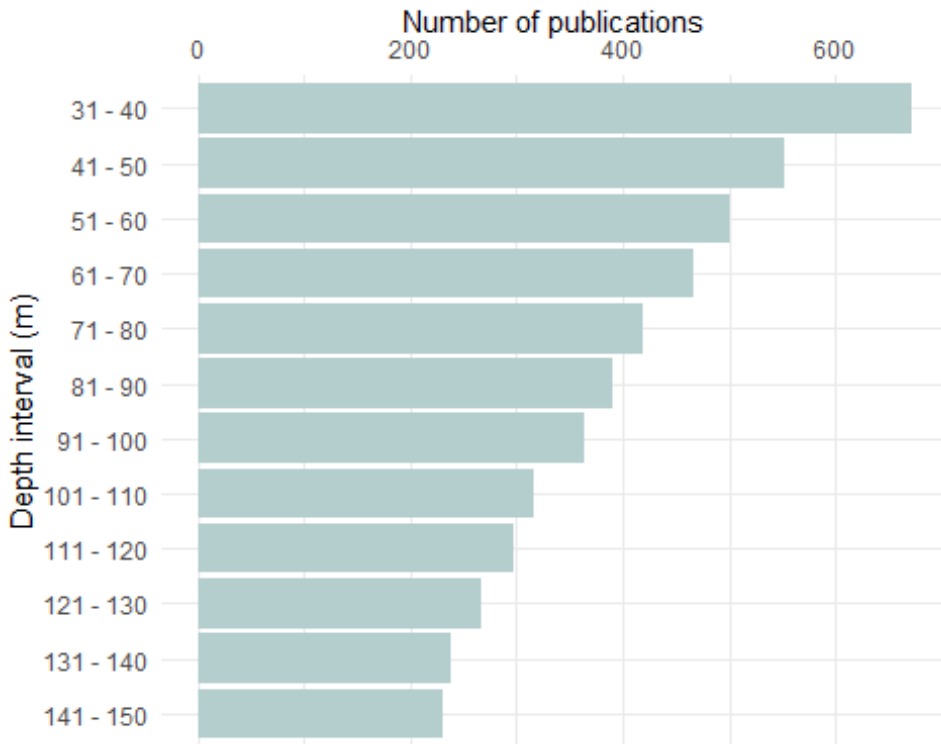
Export bar charts in a dedicated folder

```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".tiff"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis",                        "globa
l",                        "Output-Visualization","Barcharts_split_methods"),
 create.dir = T
         )
  })

lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".png"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",                        "globa
l",                        "Output-Visualization","Barcharts_split"), create.
dir = T
         )
  })
```

PUBLICATIONS PER DEPTH

Publications per depth

```
raw$mininumSurveyedDepthInMeters <- as.numeric(raw$mininumSurveyedDepthInMete
rs)

## Warning: NA introdotti per coercizione

raw$maximumSurveyedDepthInMeters <- as.numeric(raw$maximumSurveyedDepthInMete
rs)

## Warning: NA introdotti per coercizione

# Define depth bins
bins <- seq(31, 151, by = 10)

# Create a data frame to hold bin counts
pubs_depth <- data.frame(
  bin_start = bins[-length(bins)],
  bin_end = bins[-1]-1,
  count = 0
)

# For each bin, count how many studies cover that depth range
for (i in 1:nrow(pubs_depth)) {
  bin_start <- pubs_depth$bin_start[i]
  bin_end <- pubs_depth$bin_end[i]

  pubs_depth$count[i] <- sum(raw$maximumSurveyedDepthInMeters >= bin_start &
raw$mininumSurveyedDepthInMeters < bin_end, na.rm = TRUE)
}

pubs_depth$depth_bin <- paste(pubs_depth$bin_start, "-", pubs_depth$bin_end)

pubs_depth$depth_bin <- reorder(pubs_depth$depth_bin, -pubs_depth$bin_start)


ggplot(pubs_depth, aes(x = factor(depth_bin), y = count)) +
  geom_bar(stat = "identity", fill = "lightcyan3") +
  labs(x = "Depth interval (m)", y = "Number of publications") +
  theme_minimal()+
  coord_flip()+
  scale_y_continuous(position="right")
```

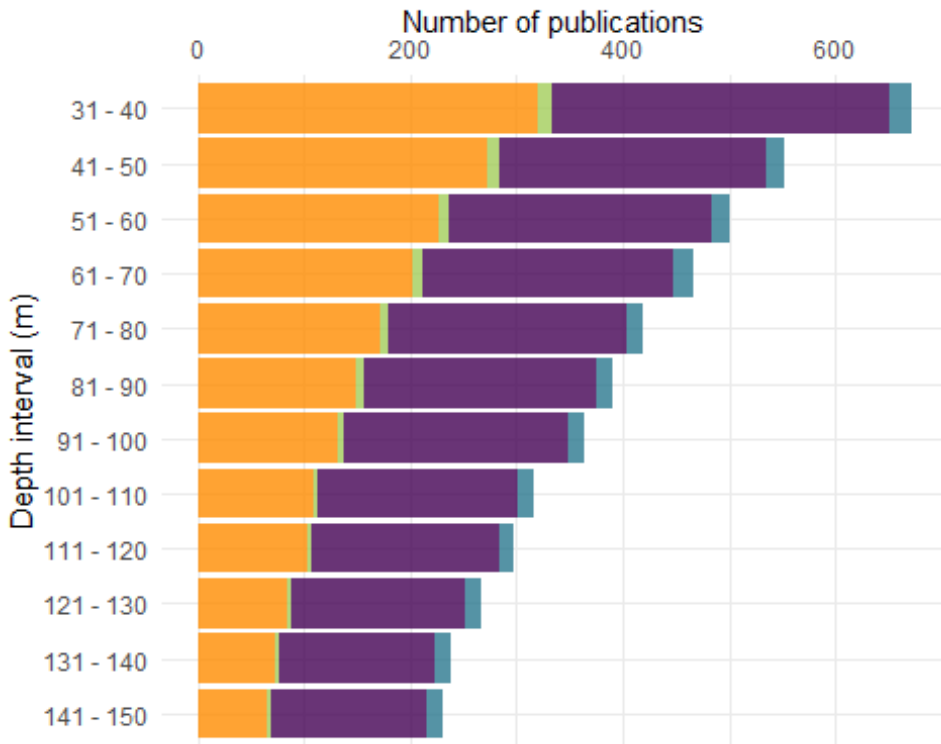


Export pubs per depth bar plot

Publication per depth by latitudinalRegion

```
# Ensure numeric values
raw$mininumSurveyedDepthInMeters <- as.numeric(raw$mininumSurveyedDepthInMete
rs)
raw$maximumSurveyedDepthInMeters <- as.numeric(raw$maximumSurveyedDepthInMete
rs)

# Define depth bins
bins <- seq(31, 150, by = 10)  # stops at 150
bin_endpoints <- bins + 9      # each bin covers 10 units (31–40, etc.)

# Create an empty results data frame
pub_depth_lat <- data.frame()

# Loop over bins and latitudinalRegions
for (i in seq_along(bins)) {
  bin_start <- bins[i]
  bin_end <- bin_endpoints[i]

  for (lat in unique(raw$latitudinalRegion)) {
    subset_data <- raw %>% filter(latitudinalRegion == lat)

    # Count how many studies cover this bin
```

```r
    count <- sum(subset_data$maximumSurveyedDepthInMeters >= bin_start &
                 subset_data$mininumSurveyedDepthInMeters < bin_end, na.rm =
TRUE)

    # Append to results
    pub_depth_lat <- rbind(pub_depth_lat, data.frame(
      depth_bin = paste(bin_start, "-", bin_end),
      latitudinalRegion = lat,
      count = count,
      bin_start = bin_start
    ))
  }
}

# Order the bins for plotting
pub_depth_lat$depth_bin <- factor(pub_depth_lat$depth_bin,
                                  levels = unique(pub_depth_lat$depth_bin[ord
er(-pub_depth_lat$bin_start)]))

# Plot
pub_depth_lat_bar <- ggplot(pub_depth_lat, aes(x = depth_bin, y = count, fill
 = latitudinalRegion)) +
  geom_bar(stat = "identity", position = "stack", alpha = 0.8) +
  labs(x = "Depth interval (m)", y = "Number of publications") +
  theme_minimal() +
  coord_flip() +
  scale_y_continuous(position = "right") +
  scale_fill_manual(values = c("#2a788e", "black", "#440154", "darkolivegreen
3", "darkorange")) +
  theme(legend.position = "none")

pub_depth_lat_bar
```



Export bar chart as pub_depth_env.png

PLATFORM BY DEPTH AND latitudinalRegion

Create a platform by depth data frame and keep meso_samplingPlatform

```
platform_depth_lat <- raw %>%
  select(File, mininumSurveyedDepthInMeters,maximumSurveyedDepthInMeters,
         latitudinalRegion, meso_direct_indirect,
         meso_samplingPlatform,
         max_depth_hov, min_depth_hov,
         max_depth_rov, min_depth_rov,
         max_depth_ccr, min_depth_ccr,
         max_depth_open, min_depth_open,
         max_depth_scuba, min_depth_scuba
         )
```

Split the platform column in one binary column per platform

```
platform_lat_lvls <-
  unique(
    unlist(
      str_split(platform_depth_lat$meso_samplingPlatform, "\\+")
                                )
                          )
```

```
platform_lat_lvls<- platform_lat_lvls[! platform_lat_lvls %in% c("",NA,"notap
p",",","nd")]

platform_depth_lat[,platform_lat_lvls] <- sapply(platform_lat_lvls,
                                        function(y){
                                          str_count(platform_depth_lat$meso_sam
plingPlatform, y)
                                          }
                                        )

rm(platform_lat_lvls)

platform_depth_lat[,-c(1:17)] <- lapply(platform_depth_lat[,-c(1:17)], as.num
eric)

platform_depth_lat[,c(1:17)] <- lapply(platform_depth_lat[,c(1:17)], as.chara
cter)


platform_depth_lat %<>%
  rename(open_direct = open,
         ccr_direct = ccr,
         scuba_direct = scuba)  %>%
  select(-meso_samplingPlatform)
```

Create a summary table with # studies per platform

```
platform_summary <- platform_depth_lat %>%
  summarise(across(where(is.numeric), ~ sum(.x, na.rm = TRUE)))

platform_summary <- as.data.frame(t(platform_summary))

platform_summary %<>%
  mutate(Platform = rownames(platform_summary)) %>%
  rename(Count = V1) %>%
  relocate(Count,.after = where(is.character))

platform_summary

##                   Platform Count
## bruv                  bruv    15
## tcs                    tcs    40
## rov                    rov   382
## trawl                trawl    11
## dredge              dredge    33
## scuba_direct scuba_direct   156
## fishery           fishery    43
## open_direct   open_direct   298
## hov                    hov   127
```

```
## grab                  grab    54
## ccr_direct      ccr_direct    91
## auv                    auv    28
## drop                  drop    37
## vams                  vams     2
## mixed                mixed     1
## drift                drift     2
## bdrop                bdrop     1
## station            station     2
## trap                   trap     2
```

Create a column with the number of platforms per study

```
platform_per_record <- platform_depth_lat %>%
  filter(meso_direct_indirect != "nd") %>%
   mutate(
     platform_per_record = rowSums(
     across(
       where(is.numeric)
       )
     )
   )

platform_per_record$platform_per_record <- as.factor(platform_per_record$plat
form_per_record)
```

Calculate number of records by number of platforms in %

```
platform_per_record <- platform_per_record %>%
  group_by(platform_per_record) %>%
  summarise(n_records = length(platform_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

platform_per_record
```

```
## # A tibble: 5 × 3
##   platform_per_record n_records percentage
##   <fct>                     <int>      <dbl>
## 1 1                           720       72.8
## 2 2                           211       21.3
## 3 3                            50        5.1
## 4 4                             7        0.7
## 5 5                             1        0.1
```

Keep most common visual platforms only

```
platform_depth_mod <- cbind(platform_depth_lat[,c(1:15)],
                            platform_depth_lat$ccr_direct,
                            platform_depth_lat$open_direct,
                            platform_depth_lat$scuba_direct,
                            platform_depth_lat$hov,
```

```
                                platform_depth_lat$rov)

names(platform_depth_mod) <- gsub("platform_depth_lat\\$", "", names(platform
_depth_mod))
```

Set max depth of studies with max depth > 150 to 150 so they fit into the plot

```
platform_depth_long <- within(platform_depth_mod, rm(mininumSurveyedDepthInMe
ters, maximumSurveyedDepthInMeters, max_depth_direct, meso_direct_indirect, h
ov, rov, open_direct, ccr_direct, scuba_direct))

platform_depth_long[,-c(1,2)] <- lapply(platform_depth_long[,-c(1,2)], as.num
eric)

str(platform_depth_long)

## 'data.frame':    994 obs. of  12 variables:
##  $ File             : chr  "AbdulWahabEtAl_2018_Biodiversity_spatial_patte
rns.md" "AbesamisEtAl_2018_Benthic_habitat_fish.md" "AchilleosEtAl_2020_Bryoz
oan_diversity_Cyprus.md" "AdamsEtAl_2020_Rhodolith_bed_discovered.md" ...
##  $ latitudinalRegion: chr  "tropical" "tropical" "temperate" "temperate"
 ...
##  $ max_depth_hov    : num  NA NA NA NA NA NA NA NA NA 100 ...
##  $ min_depth_hov    : num  NA NA NA NA NA NA NA NA NA 0 ...
##  $ max_depth_rov    : num  NA NA 477 220 100 NA 707 NA NA NA ...
##  $ min_depth_rov    : num  NA NA 118 30 30 NA 100 NA NA NA ...
##  $ max_depth_ccr    : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ min_depth_ccr    : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ max_depth_open   : num  NA NA NA NA NA NA NA NA 40 NA ...
##  $ min_depth_open   : num  NA NA NA NA NA NA NA NA 3 NA ...
##  $ max_depth_scuba  : num  NA NA NA NA NA 40 NA NA NA NA ...
##  $ min_depth_scuba  : num  NA NA NA NA NA 30 NA 1 NA NA ...

platform_depth_long  %<>%
  mutate(max_depth_scuba = case_when(max_depth_scuba >= 150 ~ 150,
                                     TRUE ~ max_depth_scuba)
         ) %>%
  mutate(max_depth_hov = case_when(max_depth_hov >= 150 ~ 150,
                                   TRUE ~ max_depth_hov)
         ) %>%
  mutate(max_depth_rov = case_when(max_depth_rov >= 150 ~ 150,
                                   TRUE ~ max_depth_rov)
         ) %>%
  mutate(max_depth_ccr = case_when(max_depth_ccr >= 150 ~ 150,
                                   TRUE ~ max_depth_ccr)) %>%
  mutate(max_depth_open = case_when(max_depth_open >= 150 ~ 150,
                                    TRUE ~ max_depth_open))
```

Set min depth of studies with min depth <30 to 30 so they fit into the plot

```
platform_depth_long %<>%
  mutate(min_depth_scuba = case_when(min_depth_scuba <= 30 ~ 30,
                                     TRUE ~ min_depth_scuba)
         ) %>%
  mutate(min_depth_hov = case_when(min_depth_hov <= 30 ~ 30,
                                     TRUE ~ min_depth_hov)
         ) %>%
  mutate(min_depth_rov = case_when(min_depth_rov <= 30 ~ 30,
                                     TRUE ~ min_depth_rov)
         ) %>%
  mutate(min_depth_ccr = case_when(min_depth_ccr <= 30 ~ 30,
                                     TRUE ~ min_depth_ccr)) %>%
  mutate(min_depth_open = case_when(min_depth_open <= 30 ~ 30,
                                     TRUE ~ min_depth_open))
```

Convert platform_depth_long to long format

```
platform_min_depth_long <- platform_depth_long %>%
  select(-c(max_depth_hov, max_depth_rov, max_depth_open, max_depth_scuba, ma
x_depth_ccr))


platform_min_depth_long[,c(1,2)] <- lapply(platform_min_depth_long[,c(1,2)],
as.factor)

platform_min_depth_long %<>%
  reshape2::melt(platform_min_depth_long,
                 id.vars = c("File","latitudinalRegion"),
                 measure.vars = grep("^min_depth_|^max_depth_",
                                     names(platform_min_depth_long),
                                     value = TRUE),
                 variable.name = "Platform",
                 value.name = "min_depth"
                 )

platform_max_depth_long <- platform_depth_long %>%
    select(-c(min_depth_hov, min_depth_rov, min_depth_open, min_depth_scuba,
min_depth_ccr))

platform_max_depth_long %<>%
    reshape2::melt(platform_max_depth_long,
                 id.vars = c("File","latitudinalRegion"),
                 measure.vars = grep("^min_depth_|^max_depth_",
                                     names(platform_max_depth_long),
                                     value = TRUE),
                 variable.name = "Platform",
                 value.name = "max_depth"
                 )
```

```r
platform_depth_lat_long <- cbind(platform_min_depth_long, platform_max_depth_
long$max_depth)

rm(platform_min_depth_long, platform_max_depth_long)

platform_depth_lat_long %<>%
  rename(max_depth =`platform_max_depth_long$max_depth`)

platform_depth_lat_long$Platform <- gsub("min_depth_", "", platform_depth_lat
_long$Platform)

platform_depth_lat_long$Platform <- gsub("max_depth_", "", platform_depth_lat
_long$Platform)
```

Set proper labelling for platform

```r
platform_depth_lat_long %<>%
  mutate(Platform = case_when(Platform == "ccr" ~ "CCR",
                              Platform == "scuba" ~ "SCUBA",
                              Platform == "open" ~ "OC",
                              Platform == "rov" ~ "ROV",
                              Platform == "hov" ~ "HOV",
                              TRUE ~ Platform)
         ) %>%
  mutate(Platform = as.factor(Platform))
```

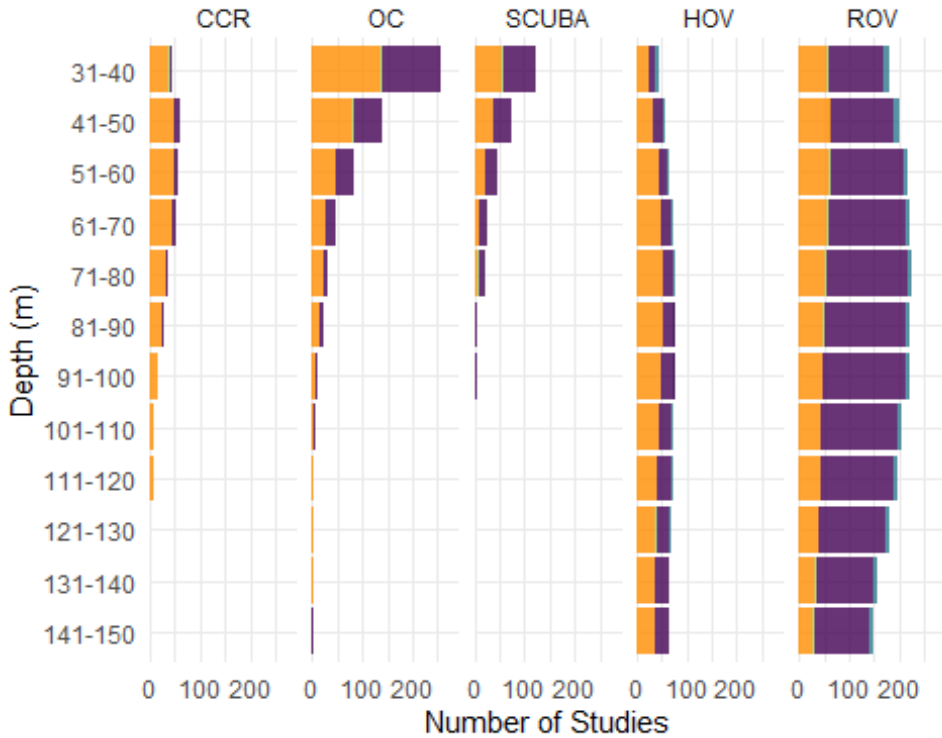
Add visual column to reorder platform

```r
platform_depth_lat_long %<>%
  mutate(Visual = as.factor(case_when(Platform == "OC" |
                                      Platform == "SCUBA" |
                                      Platform == "CCR" ~ "direct",
                              TRUE ~ "indirect" )
  )
  )

platform_depth_lat_long %<>%
  mutate(Platform = forcats::fct_reorder(Platform, as.integer(Visual)))
```

Create a bar plot with depth bins by platform and latitudinalRegion

```r
# Define 10-m bins between -30 and -150
bin_edges <- seq(-31, -151, by = -10)

# Create depth bin ranges and correctly ordered labels like "30-40"
depth_bins <- data.frame(
  bin_start = bin_edges[-length(bin_edges)],
  bin_end = bin_edges[-1]+1
) %>%
  mutate(
    bin_label = paste0(abs(bin_start), "-", abs(bin_end))  # e.g., "30-40"
```

```r
  )


# Expand each study to multiple rows for each bin it overlaps
expanded_data <- platform_depth_lat_long %>%
  rowwise() %>%
  mutate(
    bins = list(
      depth_bins %>%
        filter(bin_start > -max_depth & bin_end <= -min_depth)
    )
  ) %>%
  tidyr::unnest(bins)

# Ensure bin_label is an ordered factor from shallow to deep
expanded_data <- expanded_data %>%
  mutate(
    bin_label = factor(bin_label, levels = rev(depth_bins$bin_label))
  )

# Plot with count labels on the right
platform_depth_lat_plot <- ggplot(expanded_data, aes(x = bin_label, fill = la
titudinalRegion)) +
  geom_bar(position = "stack", alpha= 0.8) +
  scale_y_continuous(name = "Number of Studies",
                     expand = expansion(mult = c(0, 0.15))) +  # Make room fo
r text
  scale_x_discrete(name = "Depth (m)") +
  coord_flip() +  # Depth on y-axis (shallow at top)
  facet_wrap(~ Platform, nrow = 1) +
  theme_minimal() +
  labs(fill = "latitudinalRegion") +
  scale_fill_manual(values = c("#2a788e", "#440154", "darkolivegreen3", "dark
orange"))+
  theme(legend.position = "none")

platform_depth_lat_plot
```



Export bar plot of most common platform per depth interval by latitudinalRegion

```
ggsave("platform_depth_lat_plot.tiff",
       plot = platform_depth_lat_plot,
       units = "in", width = 8, height = 6, dpi = 300, compression = 'lzw',
       path = here ("Extracted_data_analysis",                        "global
",                        "Output-Visualization"))

ggsave("platform_depth_lat_plot.png",
       plot = platform_depth_lat_plot,
       units = "in", width = 8, height = 5,
       path = here ("Extracted_data_analysis", "global",
 "Output-Visualization"))
```

DIRECT - INDIRECT BY DECADE

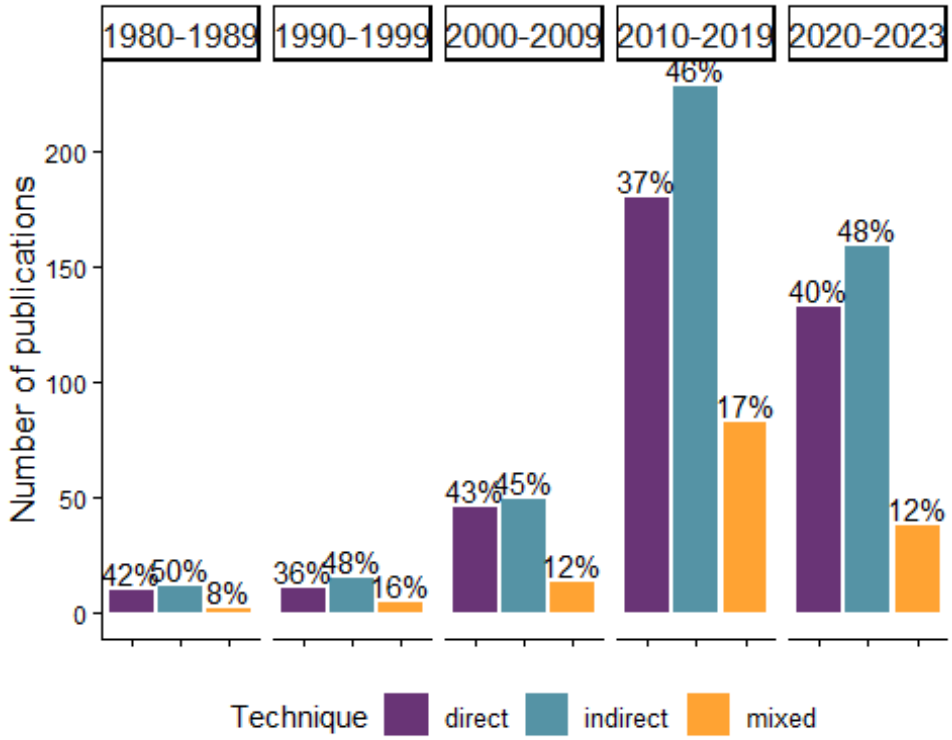
```
dir_ind_by_decade <- raw

dir_ind_by_decade %<>%
         mutate(decade = case_when(publicationYear >= 1960 & publicationYear
< 1970 ~ "1970-1979",
                                   publicationYear >= 1970 & publicationYear <
1980 ~ "1970-1979",
                                   publication Year >= 1980 & publicationYear <
1990 ~ "1980-1989",
                                   publicationYear >= 1990 & publicationYear <
```

```
2000 ~ "1990-1999",
                                  publicationYear >= 2000 & publicationYear <
2010 ~ "2000-2009",
                                  publication Year >= 2010 & publicationYear <
2020 ~ "2010-2019",
                                  publicationYear >= 2020 ~ "2020-2023")
              )

dir_ind_by_decade_lat <- dir_ind_by_decade

dir_ind_by_decade <- dir_ind_by_decade [, c("meso_direct_indirect","decade")]
 %>%
  group_by(decade,meso_direct_indirect) %>%
  tally()

colnames(dir_ind_by_decade) <- c("Decade", "Technique","Count")

unique(dir_ind_by_decade$Technique)

## [1] "direct"   "indirect" "mixed"    "nd"

dir_ind_by_decade %<>%
  filter(Technique != "nd" & Technique != "NA") %>% #remove nd from viz becau
se they accounted for a really small percentage
  group_by(Decade) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1))
```

Make bar plot of technique by decade

```
technique_decade_barplot <- ggplot(subset(dir_ind_by_decade,
              dir_ind_by_decade$Decade != "1970-1979"),
       aes(x=Technique, y=Count,
           fill=Technique, label = Technique)
       ) +
  geom_bar(position="dodge",
           stat="identity", alpha= 0.8) +
  geom_text(aes(label = paste(round(Percentage,0),"%", sep="")),
            parse = F,
            vjust = -0.25,
            size=3.9)+
  facet_grid(~ Decade,
             switch = "y") +
  theme_classic()+
  scale_fill_manual(values= c("#440154","#2a788e","darkorange")) +
  theme(legend.position = "bottom",
        legend.text = element_text(size = 10),
        axis.text.x = element_blank(),
        axis.text.y = element_text(size=8.7),
        axis.title.x = element_blank(),
        axis.title.y = element_text(size=13),
```

```
        strip.text = element_text(size = 13))+
  labs(y= "Number of publications")

technique_decade_barplot
```



Export bar plot of technique by decade

```
ggsave("technique_decade_barplot.tiff",
       plot = technique_decade_barplot,
       units = "in", width = 7, height = 6, dpi = 300, compression = 'lzw',
       path = here ("Extracted_data_analysis",                        "global
",                        "Output-Visualization"))

ggsave("technique_decade_barplot.png",
       plot = technique_decade_barplot,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                        "Output-Visualization"))
```

Technique by decade for temperate and tropical ecosystems

```
dir_ind_by_decade_lat <- dir_ind_by_decade_lat [, c("meso_direct_indirect","d
ecade","latitudinalRegion")] %>%
  group_by(decade,meso_direct_indirect, latitudinalRegion) %>%
  tally()

colnames(dir_ind_by_decade_lat) <- c("Decade", "Technique","latitudinalRegion
```

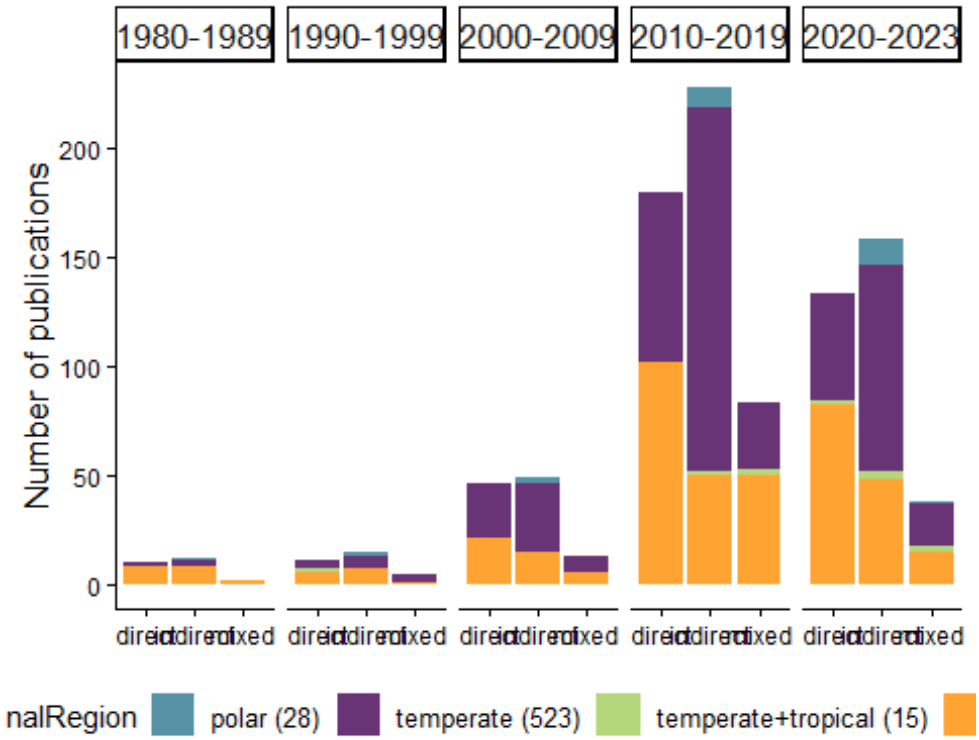
```
","Count")

unique(dir_ind_by_decade_lat$Technique)

## [1] "direct"   "indirect" "mixed"    "nd"

dir_ind_by_decade_lat %<>%
  filter(Technique != "nd" & Technique != "NA") %>% #remove nd from viz becau
se they accounted for a really small percentage
  group_by(Decade) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1))
```

Calculate publication percentage by decade and technique (not env)

```
dir_ind_by_decade_lat %<>%
  filter(Technique != "nd" & Technique != "NA") %>% #remove nd from viz becau
se they accounted for a really small percentage
  group_by(Decade) %>%
  mutate(Percentage_tech_decade = round(Count / sum(Count) * 100,1))
```

Make bar plot of technique by decade

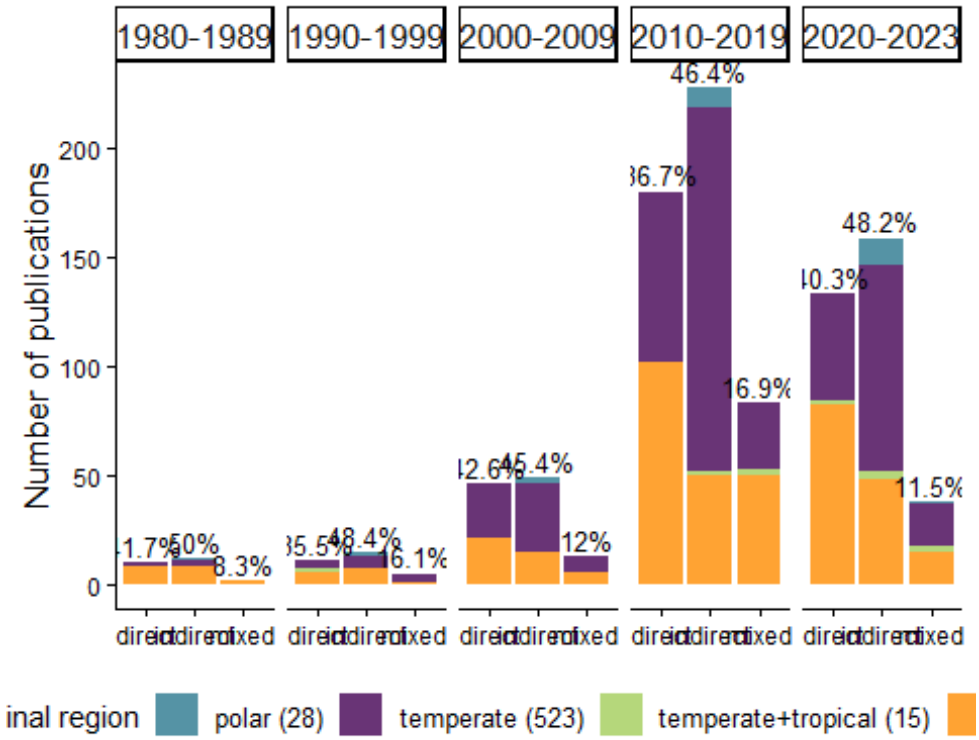
```
technique_decade__env_barplot <- ggplot(subset(dir_ind_by_decade_lat,
               dir_ind_by_decade_lat$Decade != "1970-1979" & dir_ind_by_decade
_lat$latitudinalRegion != "polar+temperate"),
       aes(x=Technique, y=Count,
           fill=latitudinalRegion, label = Technique)
       ) +
  geom_bar(position="stack",
           stat="identity", alpha= 0.8) +
  facet_grid(~ Decade,
             switch = "y") +
  theme_classic()+
  scale_fill_manual(values= c("#2a788e","#440154","darkolivegreen3","darkoran
ge"),
                    labels = c("polar (28)", "temperate (523)", "temperate+tr
opical (15)", "tropical (425)")) +
  theme(legend.position = "bottom",
        legend.text = element_text(size = 10),
        axis.text.y = element_text(size=8.7),
        axis.title.x = element_blank(),
        axis.title.y = element_text(size=13),
        strip.text = element_text(size = 13))+
  labs(y= "Number of publications")

technique_decade__env_barplot
```



Add percentage for technique and latitudinalRegion

```
# 1. Summarize label data: one row per Technique per Decade
label_data <- dir_ind_by_decade_lat %>%
  filter(Technique != "nd" & Technique != "NA") %>%
  group_by(Decade, Technique) %>%
  summarise(
    Total_Count = sum(Count),
    .groups = "drop"
  ) %>%
  group_by(Decade) %>%
  mutate(
    Percentage_label = round(Total_Count / sum(Total_Count) * 100, 1)
  )

# 2. Compute bar tops (i.e. Y position for label)
# We join with original to get full bar height, grouped by Decade and Techniq
ue
bar_tops <- dir_ind_by_decade_lat %>%
  filter(Technique != "nd" & Technique != "NA") %>%
  group_by(Decade, Technique) %>%
  summarise(
    y_position = sum(Count),
    .groups = "drop"
  )
```

```r
# 3. Merge positions with labels
label_data <- label_data %>%
  left_join(bar_tops, by = c("Decade", "Technique"))

# 4. Main bar plot data
plot_data <- dir_ind_by_decade_lat %>%
  filter(Technique != "nd" & Technique != "NA" &
         Decade != "1970-1979" & latitudinalRegion != "polar+temperate")

# 5. Filter label data for matching decades
label_data_filtered <- label_data %>%
  filter(Decade != "1970-1979")

# 6. Plot
technique_decade_env_barplot <- ggplot(plot_data,
       aes(x = Technique, y = Count, fill = latitudinalRegion)) +
  geom_bar(stat = "identity", position = "stack", alpha = 0.8) +

  # Add percentage labels above each stacked bar
  geom_text(data = label_data_filtered,
            aes(x = Technique, y = y_position,
                label = paste0(Percentage_label, "%")),
            vjust = -0.3, size = 3.5, inherit.aes = FALSE) +

  facet_grid(~ Decade, switch = "y") +
  theme_classic() +
  scale_fill_manual(
    values = c("#2a788e", "#440154", "darkolivegreen3", "darkorange"),
    labels = c("polar (28)", "temperate (523)", "temperate+tropical (15)", "t
ropical (425)")
  ) +
  theme(
    legend.position = "bottom",
    legend.text = element_text(size = 10),
    axis.text.y = element_text(size = 8.7),
    axis.title.x = element_blank(),
    axis.title.y = element_text(size = 13),
    strip.text = element_text(size = 13)
  ) +
  labs(y = "Number of publications",
       fill = "Latitudinal region")

technique_decade_env_barplot
```



Export bar plot of technique by decade

```
ggsave("technique_decade_env_barplot.tiff",
       plot = technique_decade_env_barplot,
       units = "in", width = 7, height = 6, dpi = 300, compression = 'lzw',
       path = here ("Extracted_data_analysis",                          "global
",                          "Output-Visualization"))

ggsave("technique_decade_env_barplot.png",
       plot = technique_decade_env_barplot,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                          "global
",                          "Output-Visualization"))
```

FIELD & INDICATORS

Create subset for field & indicators

```
field_merged <- raw[,c(59:length(raw))]

field_merged[,] <- lapply(field_merged[,], as.numeric)

colSums(field_merged)

##                    field_cons               field_commdesc
##                           121                          412
```

```
##                    field_map                 field_impass
##                          180                          254
##                  field_troph                   field_taxo
##                           67                          148
##                   field_dist                   field_aqua
##                          324                            4
##                 field_method                 field_physio
##                           63                          162
##                   field_func                field_habtype
##                           52                            1
##                field_fishery                  field_stock
##                           54                           57
##                  field_paleo                  field_phylo
##                            1                           48
##                   field_rest                field_archaeo
##                            9                            1
##                  field_micro            field_biomedicine
##                           10                            1
##                   field_chem     field_Molecular_ecology
##                            1                          159
##              field_Acoustics field_environmental_drivers
##                          127                          267
##              field_Behaviour         field_Geomorphology
##                          101                           55
##               field_Taxonomy
##                          126
```

Select only the most common fields

```
field_merged <- field_merged[, colSums(field_merged) > 120]

field_merged$field_Taxonomy <- NULL

colSums(field_merged)
```

```
##                   field_cons               field_commdesc
##                          121                          412
##                    field_map                 field_impass
##                          180                          254
##                   field_taxo                   field_dist
##                          148                          324
##                 field_physio     field_Molecular_ecology
##                          162                          159
##              field_Acoustics field_environmental_drivers
##                          127                          267
```

```
field_merged <- field_merged[, order(colSums(field_merged), decreasing = F)]

colSums(field_merged)
```

```
##                   field_cons              field_Acoustics
##                          121                          127
##                   field_taxo      field_Molecular_ecology
##                          148                          159
##                 field_physio                    field_map
##                          162                          180
##                 field_impass field_environmental_drivers
##                          254                          267
##                   field_dist               field_commdesc
##                          324                          412
```

Rename fields with long version names

```
field_merged %<>%
  rename(`Biodiversity and \n community structure (412)` = field_commdesc,
         `Distribution and \n connectivity (324)` = field_dist,
         `Conservation and \n management (121)` = field_cons,
         `Disturbances (254)` = field_impass,
         `Mapping (180)` = field_map,
         `Physiology (162)` = field_physio,
         `Taxonomy (148)` = field_taxo,
         `Molecular ecology (159)` = field_Molecular_ecology,
         `Acoustics (127)` = field_Acoustics,
         `Environmental drivers and \n spatio-temporal patterns (267)` = fiel
d_environmental_drivers
         )
colSums(field_merged)
```

```
##                        Conservation and \n management (121)
##                                                         121
##                                                Acoustics (127)
##                                                         127
##                                                 Taxonomy (148)
##                                                         148
##                                      Molecular ecology (159)
##                                                         159
##                                             Physiology (162)
##                                                         162
##                                                Mapping (180)
##                                                         180
##                                            Disturbances (254)
##                                                         254
## Environmental drivers and \n spatio-temporal patterns (267)
##                                                         267
##                       Distribution and \n connectivity (324)
##                                                         324
##                 Biodiversity and \n community structure (412)
##                                                         412
```

Prepare the indicator matrix

```r
ind_matrix <- split_results[["indicators"]]
```

Select only the most common indicators

```r
colSums(ind_matrix)

##        Abundance            Cover        Diversity          Habitat
##              344              309              455              180
##    Morphological       complexity             Size             taxo
##              242                6              218              121
##          Biomass           impact             surv geneticSequences
##               68               16               28              159
##          Density           growth           Health    Physiological
##              219               42               89              127
##           ALDFGs           Litter        Behaviour          Trophic
##               64               55               39               63
##         rugosity        Mortality              NIS      Recruitment
##                1               27               24               25
##             func           sample              sex         taxonomy
##               20                1               22                1
##          disease    economicValue           ecoser      orientation
##                7                1                3                1
##        Diversity        fecundity
##                1                1

ind_common_matrix <- ind_matrix[, colSums(ind_matrix) > 30]

ind_common_matrix <- ind_common_matrix[, order(colSums(ind_common_matrix), de
creasing = TRUE)]

ind_common_matrix <- as.data.frame(ind_common_matrix)

ind_common_matrix <- ind_common_matrix[!purrr::map_lgl(ind_common_matrix, ~ a
ll(is.na(.)))] # remove empty column
```

Remove taxonomy studies which have a dedicated section

```r
field_merged$`Taxonomy (148)` <- NULL
ind_common_matrix$taxo <- NULL

colSums(ind_common_matrix)

##        Diversity        Abundance            Cover    Morphological
##              455              344              309              242
##          Density             Size          Habitat geneticSequences
##              219              218              180              159
##    Physiological           Health          Biomass           ALDFGs
##              127               89               68               64
##          Trophic           Litter           growth        Behaviour
##               63               55               42               39
```

Rename indicators with long version names

```
ind_common_matrix %<>%
  rename(`Abundance \n (344)` = Abundance,
         `Cover \n (309)` = Cover,
         `Diversity \n (455)` = Diversity,
         `Habitat \n (180)` = Habitat,
         `Morphology \n (242)` = Morphological,
         `Size \n (218) ` = Size,
         `Biomass \n (68)` = Biomass,
         `Density \n (219)` = Density,
         `Growth \n rate (42)` = growth,
         `Health \n (89)`= Health,
         `Physiological \n indicator \n (127)` = Physiological,
         `ALDFGs \n (64)` = ALDFGs,
         `Litter \n (55)` = Litter,
         `Behaviour \n (39)`= Behaviour,
         `Trophic \n (63)` = Trophic,
         `Genetic \n sequence \n (159)` = geneticSequences
         )

colSums(ind_common_matrix)
```

```
##                  Diversity \n (455)                   Abundance \n (344)
##                                 455                                  344
##                      Cover \n (309)                  Morphology \n (242)
##                                 309                                  242
##                    Density \n (219)                      Size \n (218)
##                                 219                                  218
##                    Habitat \n (180)        Genetic \n sequence \n (159)
##                                 180                                  159
## Physiological \n indicator \n (127)                     Health \n (89)
##                                 127                                   89
##                     Biomass \n (68)                     ALDFGs \n (64)
##                                  68                                   64
##                     Trophic \n (63)                     Litter \n (55)
##                                  63                                   55
##                 Growth \n rate (42)                   Behaviour \n (39)
##                                  42                                   39
```

Multiply the transpose of the field matrix with the indicator matrix

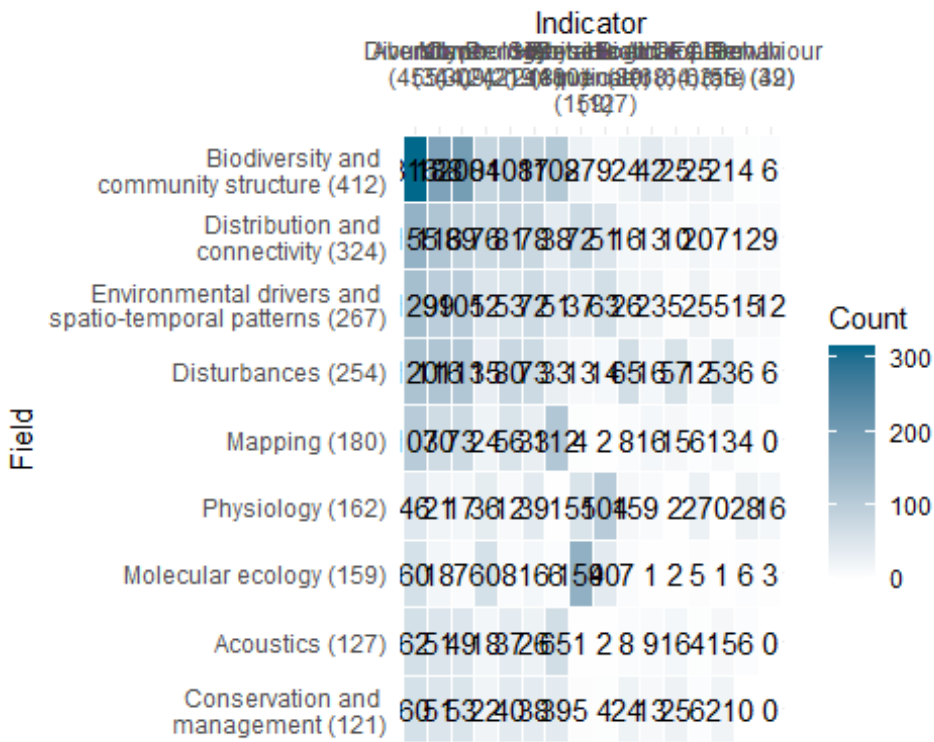
```
co_occurrence_matrix <- t(as.matrix(field_merged)) %*% as.matrix(ind_common_m
atrix)
```

Reshape the co-occurrence matrix for plotting}

```
co_occurrence_df <- reshape2::melt(co_occurrence_matrix)

colnames(co_occurrence_df) <- c("Field", "Indicator", "Count")
```

Plot the heatmap

```
field_ind_heat <- ggplot(co_occurrence_df, aes(x = Indicator, y = Field, fill
 = Count)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "deepskyblue4") +
  geom_text(aes(label = Count), color = "black") +
  theme_minimal()+
  scale_x_discrete(position="top")

field_ind_heat
```



Export plot

```
ggsave("field_ind_heat.png",
       units = "in", width = 12.5, height = 5.5,
       path = here ("Extracted_data_analysis", "global",
   "Output-Visualization"))
```

Calculate # of ecological indicators per record

Create a column with the sum of ecological indicators values to see how many different sampling units were adopted by each study

```
ind_summary <- as.data.frame(split_results[["indicators"]])

ind_summary <- ind_summary[!purrr::map_lgl(ind_summary, ~ all(is.na(.)))] # r
```

```
emove empty column

ind_summary %<>%  rowwise() %>%
   mutate(ind_per_record =  sum(c_across(where(is.numeric))))
```

Calculate number of records by number of sampling units in %

```
ind_summary$ind_per_record <- as.factor(ind_summary$ind_per_record)

ind_per_record <- ind_summary %>%
  group_by(ind_per_record) %>%
  summarise(n_records = length(ind_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

ind_per_record

## # A tibble: 10 × 3
##    ind_per_record n_records percentage
##    <fct>              <int>      <dbl>
##  1 0                      3        0.3
##  2 1                    123       12.4
##  3 2                    289       29.1
##  4 3                    279       28.1
##  5 4                    157       15.8
##  6 5                     85        8.6
##  7 6                     27        2.7
##  8 7                     24        2.4
##  9 8                      5        0.5
## 10 9                      2        0.2
```

SAMPLING UNIT

Prepare a data frame for sampling unit

```
sampling_df <- raw %>%
  filter(is.na(taxonomic_publication)) %>% # remove purely taxonomic studies
  select(File,recordTitle,DOI,publicationYear,mininumSurveyedDepthInMeters,ma
ximumSurveyedDepthInMeters,samplingUnit)

sampling_df %>% tidyr::drop_na(samplingUnit)
sampling_df[!is.na(sampling_df$samplingUnit),]

sampling_df %<>%
  filter(samplingUnit != "notapp")

sampling_df$samplingUnit <- as.character(sampling_df$samplingUnit)
```

Prepare columns per sampling units

```
sampling_lvls <- unique(unlist(str_split(sampling_df$samplingUnit,
                                          "\\+")
                               )
                        )

rownames(is.na(sampling_df$samplingUnit))

## NULL

sampling_df[,sampling_lvls] <- sapply(sampling_lvls,
                                      function(y){
                                        str_count(sampling_df$samplingUnit, y)
                                        }
                                      )

rm(sampling_lvls)

colnames(sampling_df)

##  [1] "File"                        "recordTitle"
##  [3] "DOI"                         "publicationYear"
##  [5] "mininumSurveyedDepthInMeters" "maximumSurveyedDepthInMeters"
##  [7] "samplingUnit"                "quadrat"
##  [9] "sample"                      "transect"
## [11] "expunit"                     "nd"
## [13] "permquad"                    "pit"
## [15] "permtrans"                   "lit"
## [17] "stationary"                  "permlit"
## [19] "roaming_transect"            "random_swim"
## [21] "radial_transect"             "arms"
## [23] "plotless"                    "circular_transect"
```

Calculate # of sampling unit per record

Create a column with the sum of sampling units values to see how many different sampling units were adopted by each study

```
sampling_df %<>%
   mutate(type_per_record = rowSums(across(quadrat:permlit)
                            )
   )
```

Calculate number of records by number of sampling units in %

```
sampling_df$type_per_record <- as.factor(sampling_df$type_per_record)

sampling_unit_per_record <- sampling_df %>%
  group_by(type_per_record) %>%
  summarise(n_records = length(type_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,0))
```

```
sampling_unit_per_record

## # A tibble: 5 × 3
##   type_per_record n_records percentage
##   <fct>               <int>      <dbl>
## 1 0                       1          0
## 2 1                     734         81
## 3 2                     149         16
## 4 3                      19          2
## 5 4                       1          0
```

Create a data frame with number of publication per sampling unit type

```
sampling_unit_tot <- sampling_df  %>%
  summarise(across(c(8:(length(sampling_df)-1)),
                   list(sum)))

sampling_unit_tot <- reshape2::melt(sampling_unit_tot)

## No id variables; using all as measure variables

colnames(sampling_unit_tot) <- c("Sampling_unit", "Count")

sampling_unit_tot %<>%
        mutate_at("Sampling_unit",
                  stringr::str_replace, "_1", "")
```

Simplified sampling units

Create a column with a simplified sampling units for transect and quadrat

```
  sampling_unit_tot %<>%
       mutate(Sampling_unit_macro = case_when(Sampling_unit == 'quadrat' |
                                   Sampling_unit == "permquad" ~ "Quadrat",
                                   Sampling_unit == "transect"|
                                   Sampling_unit == "lit" |
                                   Sampling_unit == "pit" |
                                   Sampling_unit == "permlit" |
                                   Sampling_unit == "permtrans" |
                                   Sampling_unit == "circular_transect" |
                                   Sampling_unit == "radial_transect"~ "Transe
ct",
                                   Sampling_unit == "nd" ~ "ND",
                                   Sampling_unit == "stationary" ~ "Stationary
",
                                   Sampling_unit == "expunit" ~ "Experimental
unit",
                                   Sampling_unit == "sample" ~ "Sample",
                                   .default = "Other")
              )
```

```r
# Change columns order
sampling_unit_tot <- cbind(sampling_unit_tot$Sampling_unit_macro,
                       sampling_unit_tot$Sampling_unit,
                       sampling_unit_tot$Count)

colnames(sampling_unit_tot) <- c("Sampling_unit_macro","Sampling_unit","Count
")

sampling_unit_tot <- as.data.frame(sampling_unit_tot)

sampling_unit_tot[,1:2] <- lapply(sampling_unit_tot[,1:2], as.factor)
sampling_unit_tot$Count <- as.numeric(sampling_unit_tot$Count)

sampling_unit_simplified <- aggregate(Count ~ Sampling_unit_macro, data = sam
pling_unit_tot, FUN = sum)

sampling_unit_simplified %<>%
  group_by(Sampling_unit_macro) %>%
  mutate(Percentage_macro= round(Count / sum(sampling_unit_simplified$Count)
* 100,1))

sampling_unit_simplified

## # A tibble: 7 × 3
## # Groups:   Sampling_unit_macro [7]
##   Sampling_unit_macro Count Percentage_macro
##   <fct>               <dbl>            <dbl>
## 1 Experimental unit      65              5.9
## 2 ND                     45              4.1
## 3 Other                  13              1.2
## 4 Quadrat               313             28.2
## 5 Sample                297             26.8
## 6 Stationary             18              1.6
## 7 Transect              358             32.3
```

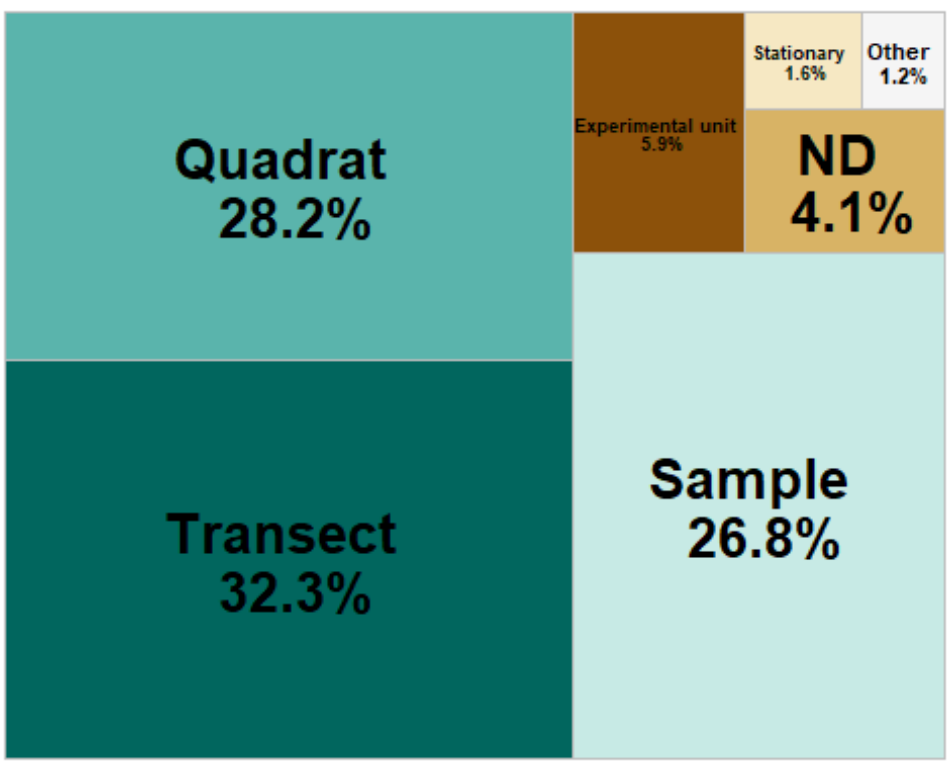
Build a treemap with simplified sampling unit

```r
library(treemapify)

RdYlBu <- RColorBrewer::brewer.pal(7, "RdYlBu")
RdYlBu

## [1] "#D73027" "#FC8D59" "#FEE090" "#FFFFBF" "#E0F3F8" "#91BFDB" "#4575B4"

sampling_simplified_treemap <-
ggplot(sampling_unit_simplified,
       aes(area=Count,
           fill=Sampling_unit_macro,
           line = "black",
```

```
            label=paste0(Sampling_unit_macro, " \n ",Percentage_macro,"%"))
        )+
  treemapify::geom_treemap(layout = "squarified")+
  geom_treemap_text(place = "centre",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values= c("#FC8D59", "#FEE090","#FFFFBF", "#E0F3F8", "#D7
3027", "#91BFDB", "#4575B4"))
  scale_fill_brewer(palette = "RdYlBu")
```

```
## <ggproto object: Class ScaleDiscrete, Scale, gg>
##     aesthetics: fill
##     axis_order: function
##     break_info: function
##     break_positions: function
##     breaks: waiver
##     call: call
##     clone: function
##     dimension: function
##     drop: TRUE
##     expand: waiver
##     fallback_palette: function
##     get_breaks: function
##     get_breaks_minor: function
##     get_labels: function
##     get_limits: function
##     get_transformation: function
##     guide: legend
##     is_discrete: function
##     is_empty: function
##     labels: waiver
##     limits: NULL
##     make_sec_title: function
##     make_title: function
##     map: function
##     map_df: function
##     minor_breaks: waiver
##     n.breaks.cache: NULL
##     na.translate: TRUE
##     na.value: NA
##     name: waiver
##     palette: function
##     palette.cache: NULL
##     position: left
##     range: environment
##     rescale: function
##     reset: function
##     train: function
##     train_df: function
```

```
##     transform: function
##     transform_df: function
##     super:  <ggproto object: Class ScaleDiscrete, Scale, gg>

sampling_unit_simplified %<>%
  mutate(case_when(Sampling_unit_macro == "Experimental unit" ~
          "Experimental \n unit", T ~ Sampling_unit_macro))

sampling_simplified_treemap <-
ggplot(sampling_unit_simplified,
       aes(area=Count,
           fill= reorder(Sampling_unit_macro, -Count),
           line = "black",
           label=paste0(Sampling_unit_macro, " \n ",Percentage_macro,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  geom_treemap_text(place = "centre", fontface = "bold",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values=c("#01665E","#5AB4AC","#C7EAE5", "#8C510A","#D8B36
5","#F6E8C3","#F5F5F5")) # selected from "BrBG" palette of RBrewer

sampling_simplified_treemap
```



Export sampling_simplified_treemap

```
ggsave("sampling_simplified_treemap.png",
       plot = sampling_simplified_treemap,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",
                    "global",
                    "Output-Visualization"))
```

Export sampling_unit_tot table for supplementary materials

```
library(openxlsx)

openxlsx::write.xlsx(sampling_unit_tot,
           file = here("Extracted_data_analysis",                        "glo
bal",                        "Output-Transformed_data",
                    "sampling_unit_tot.xlsx"))
```

Quadrat extracted from transect

```
print(paste("Number of publications where quadrat is the sampling unit or one
 of the sampling unit: ",
            nrow(quad_design <- subset(raw,grepl("quad", samplingUnit)
                                       )
                                 )
            )
)

## [1] "Number of publications where quadrat is the sampling unit or one of t
he sampling unit:  312"

print(paste("Number of publications where quadrats were collected along trans
ects: ", nrow(subset(quad_design,
            grepl("transect\\(quadrat", meso_design) |
              grepl("transect\\(depth\\(quadrat", meso_design)
            )
     )
)
)

## [1] "Number of publications where quadrats were collected along transects:
  207"
```

DATA TYPE

```
data_type <- count_results[["collectedDataType"]]

data_type$Percentage <- round(data_type$Percentage, 1)

data_type
```

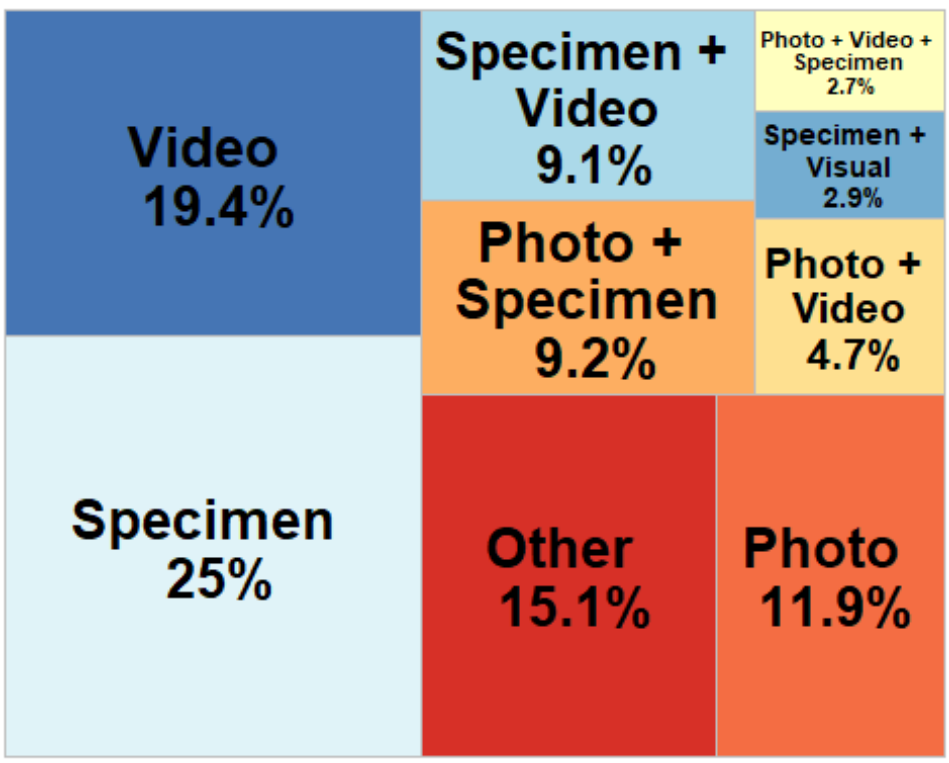
```
## # A tibble: 37 × 3
##    collectedDataType                Count Percentage
##    <fct>                            <int>      <dbl>
##  1 nd                                   1        0.1
##  2 photo                              118       11.9
##  3 photo+sediment+video                 2        0.2
##  4 photo+specimen                      91        9.2
##  5 photo+specimen+sediment              1        0.1
##  6 photo+specimen+sediment+video        3        0.3
##  7 photo+specimen+sediment+visual       2        0.2
##  8 photo+specimen+video                27        2.7
##  9 photo+specimen+video+visual          6        0.6
## 10 photo+specimen+video+water           2        0.2
## # ℹ 27 more rows
```

Lump uncommon combination into “Other”

```
data_type_simplified <- data_type %>%
  rename(`Data type`= collectedDataType) %>%
  mutate(`Data type` = as.character(`Data type`)) %>%
  filter(Percentage > 2) %>%
  bind_rows(
    data_type %>%
      filter(Percentage <= 2) %>%
      summarise(
        `Data type` = "other",
        Count = sum(.data$Count),
        Percentage = sum(.data$Percentage)
      )
  )


data_type_simplified %<>%
       mutate(`Data type` =
                case_when(`Data type` == 'photo'  ~ "Photo",
                          `Data type` == "video"~ "Video",
                          `Data type` == "other" ~ "Other",
                          `Data type` == "specimen" ~ "Specimen",
                          `Data type` == "photo+specimen" ~ "Photo + \n Speci
men",
                          `Data type` == "photo+specimen+video" ~ "Photo + Vi
deo + \n Specimen",
                          `Data type` == "photo+video" ~ "Photo + \n Video",
                          `Data type` == "photo+visual" ~ "Photo + \n Visual",
                          `Data type` == "specimen+visual" ~ "Specimen + \n V
isual",
                          `Data type` == "specimen+video" ~ "Specimen + \n Vi
deo",
                                  .default = "Other")
               )
```

Build a treemap with data type

```
library(treemapify)
library(RColorBrewer)

data_type_treemap <-
ggplot(data_type_simplified,
       aes(area=Count,
           fill=`Data type`,
           line = "black",
           label=paste0(`Data type`, " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  geom_treemap_text(place = "centre", fontface = "bold",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
    scale_fill_brewer(palette = "RdYlBu")

data_type_treemap
```



Export sampling_simplified_treemap

```
ggsave("data_type_treemap.png",
       plot = data_type_treemap,
       units = "in", width = 7, height = 6,
```

```
       path = here ("Extracted_data_analysis",                        "global
",                        "Output-Visualization"))
```

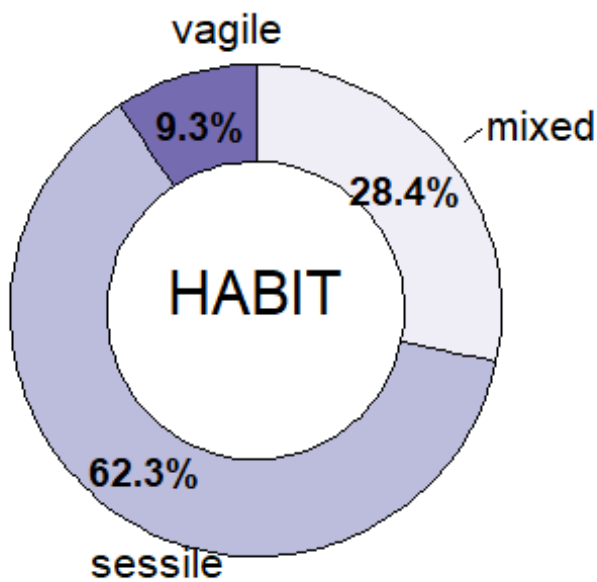
HABIT

```
habit <- as.data.frame(count_results["habit"])

# Rename columns
habit %<>%
  rename(Habit = habit.habit,
         Count = habit.Count,
         Percentage = habit.Percentage)
```

Make the habit donut plot

```
habit %<>%
  mutate(ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )

# Split and assign labels
habit_inside <- habit %>%
  mutate(label = paste0(round(Percentage,1), "%"))

habit_outside <- habit %>%
  mutate(label = Habit)

# Plot
habit_donut <- ggplot() +
  # Donut segments
  geom_rect(data = habit,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Habit),
            color = "black") +

  # Inside labels
  geom_text(data = habit_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with Habit name
  ggrepel::geom_text_repel(data = habit_outside,
                  aes(x = 4.3, y = mid, label = label),
                  size = 5.5,
                  nudge_x = 0.5,
                  direction = "y",
                  segment.color = "black",
```

```
                  hjust = 0,
                  box.padding = 0.3,
                  max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "Purples") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("HABIT"), size = 8)

habit_donut
```



Export habit_donut

```
ggsave("habit_donut.png",
       plot = habit_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                          "Output-Visualization"))
```

SUBSTRATE

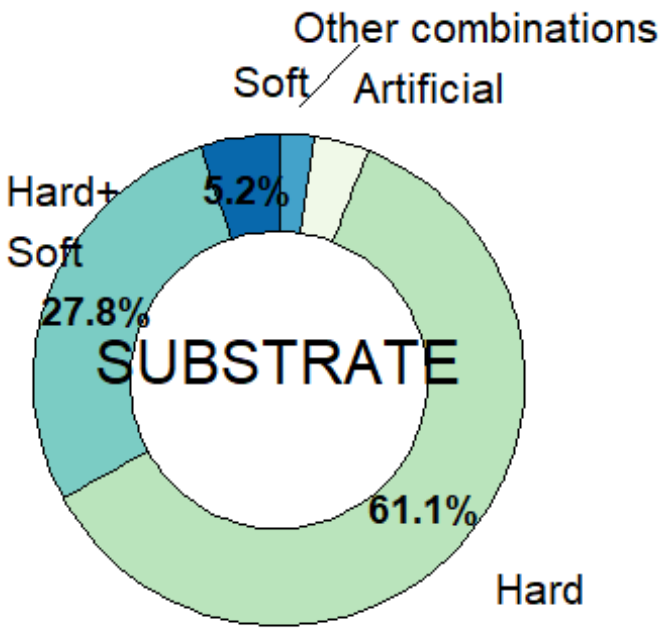
```
substrate <- as.data.frame(count_results["meso_substrate"])

# Rename columns
```

```
substrate %<>%
  rename(Substrate = meso_substrate.meso_substrate,
         Count = meso_substrate.Count,
         Percentage = meso_substrate.Percentage)

# Remove rows with nd
substrate <- substrate[!substrate$Substrate %in% "nd",]

# Merge substrate combination with less than 3% and recalculate Count and Per
centages
substrate <- substrate %>%
  mutate(Substrate = if_else(Percentage < 3, "Other combinations", Substrate))
 %>%
  group_by(Substrate) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )

substrate %<>%
  mutate(
    Substrate = case_when(Substrate == "hard" ~ "Hard",
                          Substrate == "soft"~ "Soft",
                          Substrate == "mixed" ~ paste0("Hard+","\n","Soft"),
                          Substrate == "artificial" ~ "Artificial",
                          Substrate == "Other combinations" ~ "Other combinat
ions"))
```

Make the substrate donut plot

```
substrate %<>%
  mutate(
    ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )


# Split and assign labels
substrate_inside <- substrate %>%
    filter(Percentage >= 5) %>%
  mutate(label = paste0(Percentage, "%"))

substrate_outside <- substrate %>%
  mutate(label = Substrate)

# Plot
```

```r
substrate_donut <- ggplot() +
  # Donut segments
  geom_rect(data = substrate,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Substr
ate),
            color = "black") +

  # Inside labels
  geom_text(data = substrate_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with substrate name
  ggrepel::geom_text_repel(data = substrate_outside,
                 aes(x = 4.4, y = mid, label = label),
                 size = 5.5,
                 nudge_x = 0.3,
                 direction = "y",
                 segment.color = "black",
                 hjust = 0,
                 box.padding = 0.3,
                 max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "GnBu") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("SUBSTRATE"), size = 8)

substrate_donut
```



Export the substrate donut plot

```
ggsave("substrate_donut.png",
       plot = substrate_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                          "Output-Visualization"))
```

BENTHIC POSITION

Create a data frame with benthic position

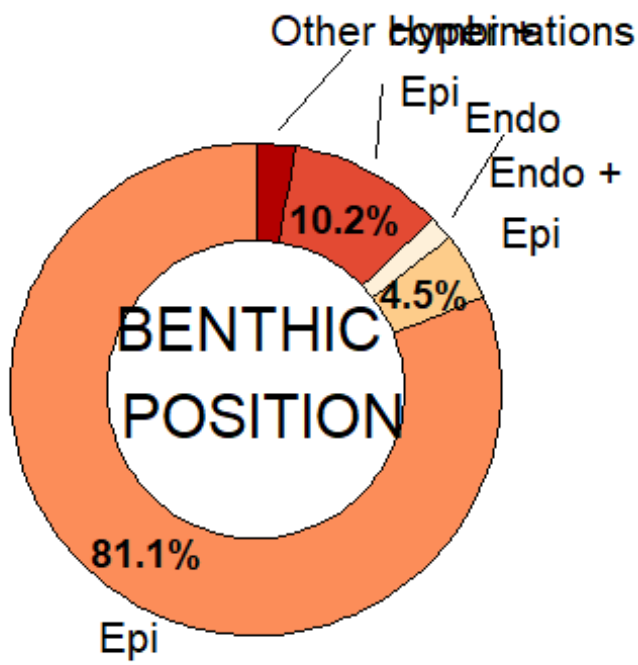
```
position <- as.data.frame(count_results["environmentalPosition"])

# Rename columns
position %<>%
  rename(Position = environmentalPosition.environmentalPosition,
         Count = environmentalPosition.Count,
         Percentage = environmentalPosition.Percentage)

# Merge position combination with less than 3% and recalculate Count and Perc
entages
position <- position %>%
  mutate(Position = if_else(Percentage < 1, "Other combinations", Position))
 %>%
  group_by(Position) %>%
```

```
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )
```

Make the benthic position donut plot

```
position %<>%
  mutate(
    Position = case_when(Position == "epi" ~ "Epi",
                         Position == "endo"~ "Endo",
                         Position == "dem+epi" ~ "Hyper + \n Epi",
                         Position == "endo+epi" ~ "Endo + \n Epi",
                         Position == "Other combinations" ~ "Other combinati
ons"),
    ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )


# Split and assign labels
position_inside <- position %>%
  filter(Percentage > 3) %>%
  mutate(label = paste0(Percentage, "%"))

position_outside <- position %>%
  mutate(label = Position)

# Plot
benthic_position_donut <- ggplot() +
  # Donut segments
  geom_rect(data = position,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Positi
on),
            color = "black") +

  # Inside labels
  geom_text(data = position_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with Position name
  ggrepel::geom_text_repel(data = position_outside,
                   aes(x = 4.2, y = mid, label = label),
                   size = 5.5,
```

```
                  nudge_x = 0.3,
                  direction = "y",
                  segment.color = "black",
                  hjust = 0,
                  box.padding = 0.3,
                  max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "OrRd") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("BENTHIC \n POSITION"), size = 8)

benthic_position_donut
```



Export the benthic_position donut plot

```
ggsave("benthic_position_donut.png",
       plot = benthic_position_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                          "Output-Visualization"))
```

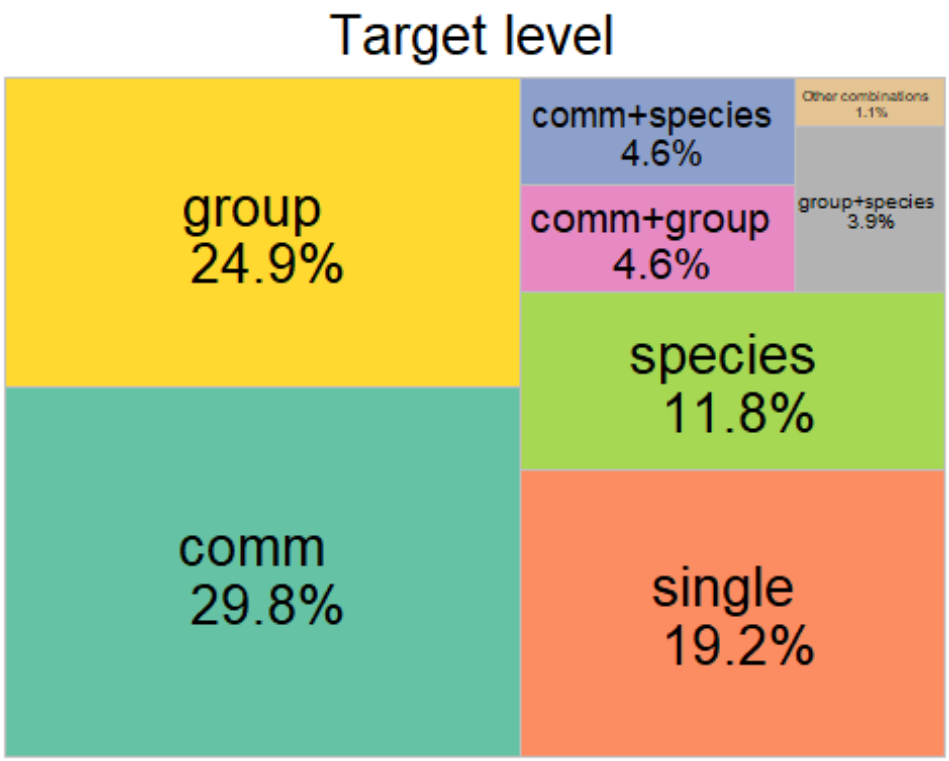
TARGET

Target level

```
target_level <- as.data.frame(count_results["taxonomicTargetLevel"])

# Rename columns
target_level %<>%
  rename(Level = taxonomicTargetLevel.taxonomicTargetLevel,
         Count = taxonomicTargetLevel.Count,
         Percentage = taxonomicTargetLevel.Percentage)

# Remove rows with nd
target_level <- target_level[!target_level$Level %in% "species,species",]
target_level <- target_level[!target_level$Level %in% ",species",]

# Merge habit combination with less than 3% and recalculate Count and Percent
ages
target_level <- target_level %>%
  mutate(Level = if_else(Percentage < 3, "Other combinations", Level)) %>%
  group_by(Level) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )
```

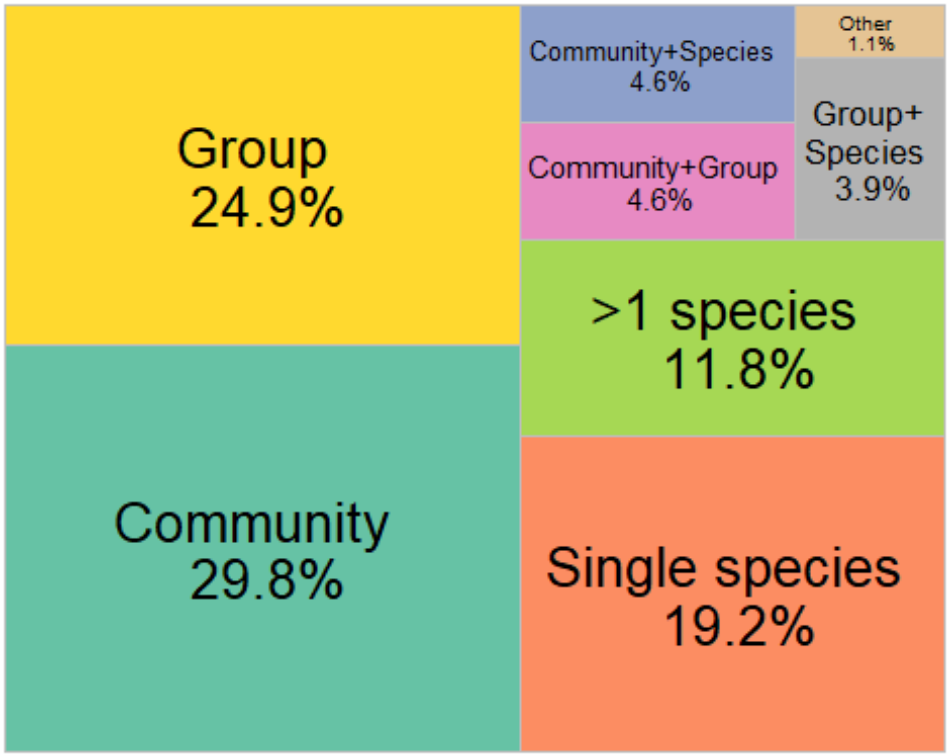
Treemap with the taxonomic target level

```
ggplot(target_level,
       aes(area=Count,
           fill=Level,
           line = "black",
           label=paste0(Level,
                        " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                    size = 20)+
  labs(title = "Target level")+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values= c("#66C2A5","#E78AC3" , "#8DA0CB","#FFD92F","#B3B
3B3", "#E5C494","#FC8D62","#A6D854"))
```




```
target_level_labels <- target_level %<>%
  mutate(
    Level = case_when(Level == "comm" ~ "Community",
                      Level == "group"~ "Group",
                      Level == "single" ~ "Single species",
                      Level == "species" ~ ">1 species",
                      Level == "comm+group" ~ "Community+Group",
                      Level == "comm+species" ~ "Community+Species",
                      Level == "group+species" ~ "Group+\nSpecies",
                      Level == "Other combinations" ~ "Other")
    )

target_level_treemap <- ggplot(target_level_labels,
       aes(area=Count,
           fill=Level,
           line = "black",
           label=paste0(Level,
                        " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values= c("#A6D854" ,"#66C2A5","#E78AC3", "#8DA0CB","#FFD
92F","#B3B3B3", "#E5C494", "#FC8D62"))
```

```
target_level_treemap
```




```
ggsave("target_level_treemap.png",
       plot = target_level_treemap,
       units = "in", width = 9, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                          "Output-Visualization"))
```

Detected phyla in community studies

```
detected <- raw %>%
    filter(str_detect(taxonomicTargetLevel,"comm")) %>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
         detected_meso_echino,
         detected_meso_other_taxa)

detected_lvls <-
  unique(
```

```
    unlist(
      str_split(detected$detected_meso_other_taxa, "\\+")
                                 )
                           )
detected_lvls<- detected_lvls[! detected_lvls %in% c("",NA,"notapp",",","nd")]
detected[,detected_lvls] <- sapply(detected_lvls,
                                       function(y){
                                         str_count(detected$detected_meso_othe
r_taxa, y)
                                         }
                                       )

rm(detected_lvls)

detected %<>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x)))

detected <- detected %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x", "x,", "xx") ~ "1",
    .x == "," ~ "0",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))
```

Calculate number of publications per taxonomic group

```
detected_tot <-
  detected %>%
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

detected_tot  <- reshape2::melt(detected_tot, id.vars = NULL)

colnames(detected_tot) <- c("Taxa", "Count")

detected_tot %<>%
        mutate_at("Taxa",
                  stringr::str_replace, "_1", "") %>%
        mutate_at("Taxa",
                  stringr::str_replace, "detected_meso_", "")


detected_tot %<>%
```

```
      mutate(Percentage= round(Count / nrow(detected) * 100,1))

rm(detected)
```

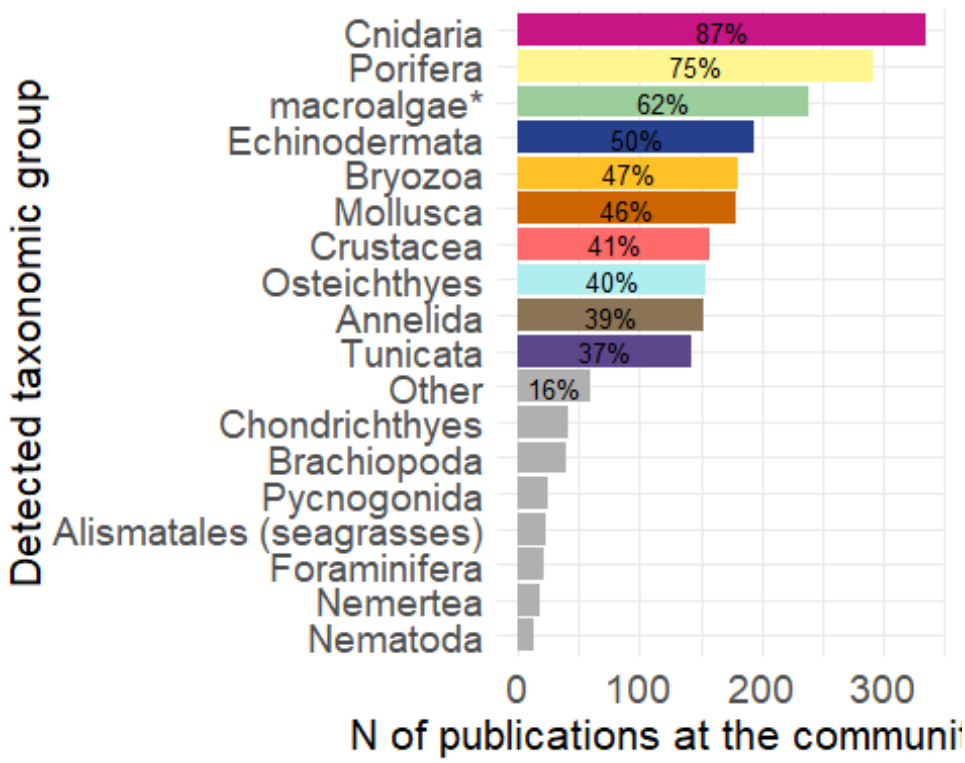
Merge poorly represented taxonomic groups into “Other”

```
detected_tot_collapsed <- detected_tot %>%
  mutate(Taxa = if_else(Percentage < 3, "Other", Taxa)) %>%
  group_by(Taxa) %>%
  summarise(
    Count = sum(Count),
    Percentage = sum(Percentage)
  )
```

Create a bar chart with detected taxonomic groups

```
detected_bar <- ggplot(detected_tot_collapsed, aes(x = reorder(Taxa, Count),
y = Count, fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(
    data = detected_tot_collapsed %>% filter(Percentage > 11),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 3.5
  )+
  labs(y = "N of publications at the community level", x = "Detected taxonomi
c group") +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")+
  scale_x_discrete(labels = rev(c("Cnidaria",
                                  "Porifera",
                                  "macroalgae*",
                                  "Echinodermata",
                                  "Bryozoa",
                                  "Mollusca",
                                  "Crustacea",
                                  "Osteichthyes",
                                  "Annelida",
                                "Tunicata",
                                "Other",
                                "Chondrichthyes",
                                "Brachiopoda",
                                "Pycnogonida",
                                "Alismatales (seagrasses)",
                                "Foraminifera",
                                "Nemertea",
                                "Nematoda")))
```

```r
detected_palette <- c("algae" ="darkseagreen3",
                      "cnidaria" = "mediumvioletred",
                      "crustacea" = "indianred1",
                      "sponges" = "khaki1",
                      "mollusca" = "darkorange3",
                      "fish" = "paleturquoise2",
                      "echino" = "royalblue4",
                      "Tunicata" = "mediumpurple4",
                      "chondrichthyes" = "gray69",
                      "Brachiopoda" = "gray69",
                      "Other" = "gray69",
                      "Nemertea" = "gray69",
                      "Pycnogonida" = "gray69",
                      "seagrass" =  "gray69",
                      "Nematoda" = "gray69",
                      "Foraminifera"   =    "gray69",
                      "Ciliata" = "gray69",
                      "Bryozoa" = "goldenrod1",
                      "Annelida" = "burlywood4")

detected_bar +
  scale_fill_manual(values = detected_palette,
    limits = names(detected_palette))+
  theme(axis.text = element_text(size = 14),
        axis.title = element_text(size = 16)
        )
```



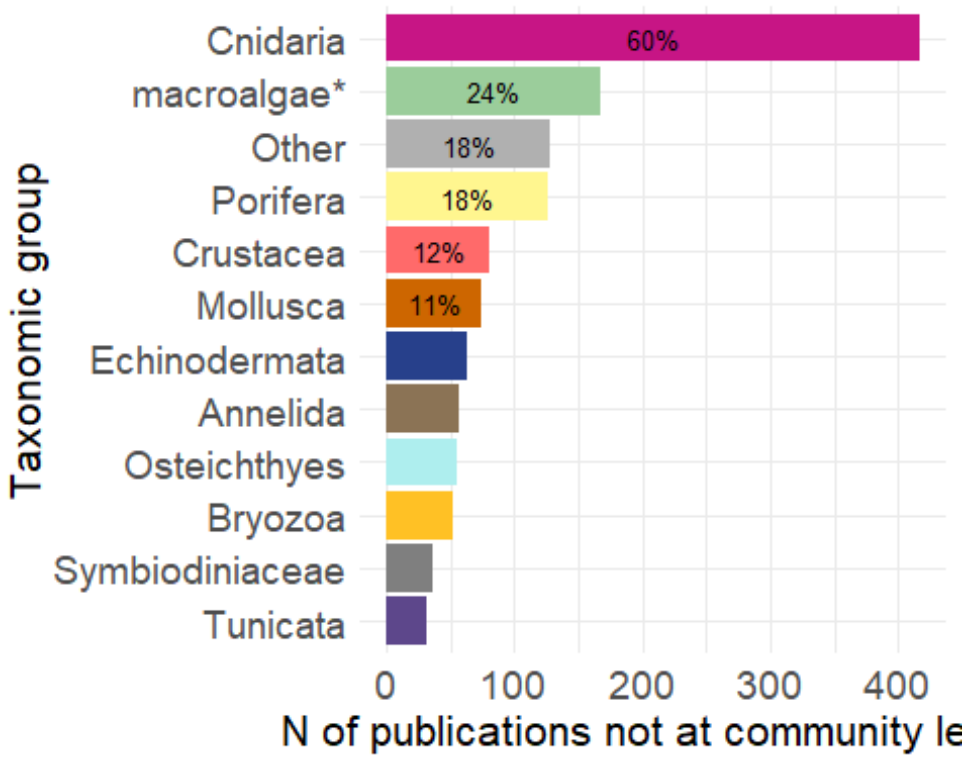
```
ggsave("detected_bar.png",
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                          "Output-Visualization"))
```

Target taxonomic groups not at the community level

```
group <- raw %>%
    filter(taxonomicTargetLevel != "comm") %>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
         detected_meso_echino,
         detected_meso_other_taxa,
         targetTaxa)

group_lvls <-
  unique(
    unlist(
      str_split(group$detected_meso_other_taxa, "\\+")
                                 )
                          )
group_lvls<- group_lvls[! group_lvls %in% c("",NA,"notapp",",","nd")]
group[,group_lvls] <- sapply(group_lvls,
                                        function(y){
                                          str_count(group$detected_meso_other_t
axa, y)
                                          }
                                        )

rm(group_lvls)

group <- group %>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa,-targetTaxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x))) %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x", "x,", "xx") ~ "1",
    .x == "," ~ "0",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))
```

```r
group_tot <-
  group %>%
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

group_tot  <- reshape2::melt(group_tot, id.vars = NULL)

colnames(group_tot) <- c("Taxa", "Count")

group_tot %<>%
        mutate_at("Taxa",
                  stringr::str_replace, "_1", "") %>%
        mutate_at("Taxa",
                  stringr::str_replace, "detected_meso_", "")


group_tot %<>%
      mutate(Percentage= round(Count / nrow(group) * 100,1))

group_tot_collapsed <- group_tot %>%
  mutate(Taxa = if_else(Percentage < 3, "Other", Taxa)) %>%
  group_by(Taxa) %>%
  summarise(
    Count = sum(Count),
    Percentage = sum(Percentage),
    .groups = "drop"
  )

group_tot_collapsed %<>%
  mutate(Taxa = case_when(Taxa == 'echino' ~ "Echinodermata",
                          Taxa == 'fish' ~ "Osteichthyes",
                          Taxa == 'mollusca' ~ "Mollusca",
                          Taxa == 'seagrass' ~ "Alismatales (seagrasses)",
                          Taxa == "chondrichthyes" ~ "Chondrichthyes",
                          Taxa == "sponges" ~ "Porifera",
                          Taxa == "crustacea" ~ "Crustacea",
                          Taxa == "cnidaria" ~ "Cnidaria",
                          Taxa == "algae" ~ "macroalgae*",
         .default = Taxa))

group_bar <- ggplot(group_tot_collapsed, aes(x = reorder(Taxa, Count), y = Co
unt, fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(
    data = group_tot_collapsed %>% filter(Percentage > 10),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
```

```
    size = 3.5
  )+
  labs(y = "N of publications not at community level", x = "Taxonomic group")
 +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")

group_palette <- c("macroalgae*" ="darkseagreen3",
                      "Cnidaria" = "mediumvioletred",
                      "Crustacea" = "indianred1",
                      "Porifera" = "khaki1",
                      "Mollusca" = "darkorange3",
                      "Osteichthyes" = "paleturquoise2",
                      "Echinodermata" = "royalblue4",
                      "Tunicata" = "mediumpurple4",
                      "Chondrichthyes" = "gray69",
                      "Other" = "gray69",
                      "Alismatales (seagrasses)" =  "gray69",
                      "Bryozoa" = "goldenrod1",
                      "Annelida" = "burlywood4")

group_bar <- group_bar +
  scale_fill_manual(values = group_palette,
    limits = names(group_palette))+
    theme(axis.text = element_text(size = 14),
        axis.title = element_text(size = 16)
        )

group_bar
```



Export bar plot

```
ggsave("group_bar.png",
       plot = group_bar,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                    "global
",                          "Output-Visualization"))
```

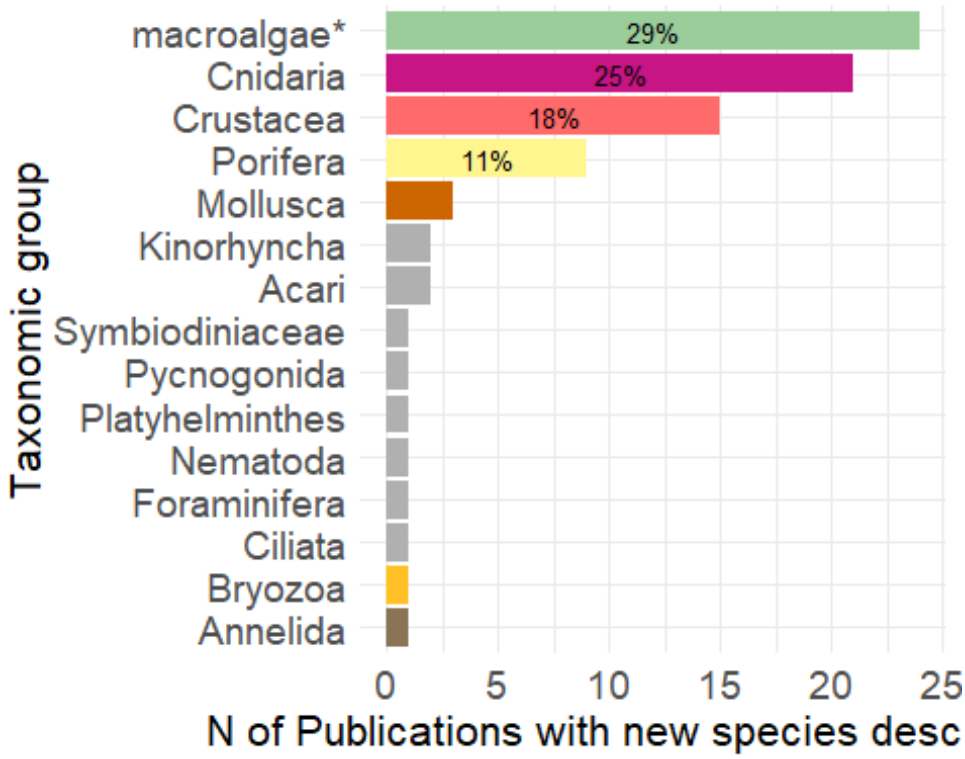
New species described per phyla

```
new_spec <- raw %>%
  filter(str_detect(taxonomic_publication, "taxa"))

new_spec %<>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
         detected_meso_echino,
         detected_meso_other_taxa)

new_spec_lvls <-
  unique(
    unlist(
```

```
      str_split(new_spec$detected_meso_other_taxa, "\\+")
                                )
                          )
new_spec_lvls<- new_spec_lvls[! new_spec_lvls %in% c("",NA,"notapp")]
new_spec[,new_spec_lvls] <- sapply(new_spec_lvls,
                                   function(y){
                                     str_count(new_spec$detected_meso_othe
r_taxa, y)
                                     }
                                   )

rm(new_spec_lvls)

new_spec  %<>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x))) %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x") ~ "1",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))

new_spec_tot <-
  new_spec %>%
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

rm(new_spec)
new_spec_tot  <- reshape2::melt(new_spec_tot, id.vars = NULL)

colnames(new_spec_tot) <- c("Taxa", "Count")

new_spec_tot %<>%
        mutate_at("Taxa",
                  stringr::str_replace, "_1", "") %>%
          mutate_at("Taxa",
                  stringr::str_replace, "detected_meso_", "")


new_spec_tot %<>%
  filter(Count>0) %>%
      mutate(Percentage= round(Count / sum(Count) * 100,1))

new_spec_plot <-   ggplot(new_spec_tot,
                          aes(x = reorder(Taxa, Count), y = Count,
                              fill= Taxa)) +
```

```
  geom_bar(stat = "identity") +
  geom_text(
    data = new_spec_tot %>% filter(Percentage > 8),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 3.5
  )+
  labs(y = "N of Publications with new species descriptions", x = "Taxonomic
group") +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")+
  scale_x_discrete(labels = rev(c("macroalgae*",
                                  "Cnidaria",
                                  "Crustacea",
                                  "Porifera",
                                  "Mollusca",
                                  "Kinorhyncha",
                                  "Acari",
                                  "Symbiodiniaceae",
                              "Pycnogonida",
                              "Platyhelminthes",
                              "Nematoda",
                              "Foraminifera",
                              "Ciliata",
                              "Bryozoa",
                              "Annelida"))
                    )

new_spec_palette <- c("algae" ="darkseagreen3",
                      "cnidaria" = "mediumvioletred",
                      "crustacea" = "indianred1",
                      "sponges" = "khaki1",
                      "mollusca" = "darkorange3",
                      "Kinorhyncha" = "gray69",
                      "Acari" = "gray69",
                      "Symbiodiniaceae" = "gray69",
                      "Pycnogonida" = "gray69",
                      "Platyhelminthes" =  "gray69",
                      "Nematoda" = "gray69",
                      "Foraminifera"   =    "gray69",
                      "Ciliata" = "gray69",
                      "Bryozoa" = "goldenrod1",
                      "Annelida" = "burlywood4")

new_species_bar <- new_spec_plot +
  scale_fill_manual(values = new_spec_palette,
    limits = names(new_spec_palette))+
    theme(axis.text = element_text(size = 14),
```

```
        axis.title = element_text(size = 16)
        )

new_species_bar
```



Export bar plot

```
ggsave("new_species_bar.png",
       plot = new_species_bar,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "global
",                          "Output-Visualization"))
```

New species descriptions that included molecular analyses

```
new_spec_molecular <- raw %>%
  filter(str_detect(taxonomic_publication, "taxa") &
           molecularAnalyses != "no" &
           molecularAnalyses != "nd" &
           molecularAnalyses != "notapp")

nrow(new_spec_molecular)

## [1] 34
```

DATA CROSSING

```
table(raw$meso_protection)

##
## mixed    nd    no   yes
##   108   652    72   162
```

VISUALLY SURVEYED SURFACE

```
raw$visuallySurveyedSurface <- as.factor(raw$visuallySurveyedSurface)
summary(subset(raw, is.na(raw$taxonomic_publication))$visuallySurveyedSurface)

##           <1           >1          >10         >100        >1000       >10
000
##           11           28           78          127          118
 63
##      >100000     >1000000    >10000000   >100000000 >10000000000
 nd
##           24            8            1            1            1
352
##           no       notapp
##            2          105
```

CCR and latitudinalRegion

```
nrow(subset(platform_depth_lat,platform_depth_lat$ccr_direct == 1))

## [1] 91

nrow(subset(platform_depth_lat,platform_depth_lat$ccr_direct == 1 & grepl("tr
opical",platform_depth_lat$latitudinalRegion)))

## [1] 80

nrow(subset(platform_depth_lat,platform_depth_lat$ccr_direct == 1 & grepl("te
mperate",platform_depth_lat$latitudinalRegion)))

## [1] 12

nrow(subset(platform_depth_lat,platform_depth_lat$hov == 1 & grepl("tropical",
platform_depth_lat$latitudinalRegion)))

## [1] 85

nrow(subset(platform_depth_lat,platform_depth_lat$hov == 1 & grepl("temperate
",platform_depth_lat$latitudinalRegion)))

## [1] 38

nrow(subset(platform_depth_lat,platform_depth_lat$hov == 1))

## [1] 127
```

```
nrow(subset(platform_depth_lat,platform_depth_lat$hov == 1 & grepl("polar",pl
atform_depth_lat$latitudinalRegion)))
```

```
## [1] 6
```

Observation

```
print(paste("Number of manipulative experimental studies:",nrow(subset(raw, g
repl("exp", raw$observationalORexperimental)))))
```

```
## [1] "Number of manipulative experimental studies: 97"
```

```
print(paste("Number of manipulative experimental studies in the field of rest
oration:", nrow(subset(raw, grepl("exp", raw$observationalORexperimental) & r
aw$field_rest != 0)

               )
            )
      )
```

```
## [1] "Number of manipulative experimental studies in the field of restorati
on: 6"
```

Temp replicates & exp

```
print(paste("# Publications that included temporal replicates:", nrow(subset
(raw, grepl("yes", temporalReplicates)))))
```

```
## [1] "# Publications that included temporal replicates: 242"
```

```
print(paste("# Publications that included temporal replicates:", nrow(subset
(raw, (grepl("exp", raw$observationalORexperimental) & grepl("yes", temporalR
eplicates))))))
```

```
## [1] "# Publications that included temporal replicates: 68"
```

Health

```
health <- subset(raw, grepl("Health", indicators) | grepl("Disease", indicato
rs))

unique(health$targetTaxa)
```

```
##  [1] "Corallinales"

##  [2] "no"

##  [3] "Paramuricea_clavata"

##  [4] "Corallium_rubrum"

##  [5] "Bryozoa+Corallium_rubrum+Leptopsammia_pruvoti+Madracis_pharensis"
```

```
##  [6] "Eunicella_cavolini+Paramuricea_clavata"

##  [7] "Pachastrella monilifera+Poecillastra_compressa"

##  [8] "Peltaster_placenta+Peltaster_placenta"

##  [9] "Agaricia_grahamae+Agaricia_undata+Madracis_pharensis"

## [10] "Scleractinia"

## [11] "Eunicella_verrucosa"

## [12] "Corallium_rubrum+Eunicella_cavolini"

## [13] "Adeonella_calveti+Myriapora_truncata+Pentapora_fascialis+Reteporella
_grimaldii+Smittina_cervicornis"
## [14] "Eunicella_cavolini+Eunicella singularis+Paramuricea_clavata"

## [15] "Cidaris_spp+Corallium_rubrum+Eunicella_cavolinii"

## [16] "Anthipathella_subpinnata+Eunicella_cavolini+Paramuricea_clavata"

## [17] "Viminella_flagellum"

## [18] "Eunicella_singularis"

## [19] "Porifera"

## [20] "Octocorallia"

## [21] "Anthozoa"

## [22] "notapp"

## [23] "Alveopora_allingi+Alveopora_ocellata"

## [24] "Eunicella_cavolinii+Eunicella_singularis+Eunicella_verrucosa+Paramur
icea_clavata"
## [25] "Placopecten_magellanicus"

## [26] "Xestospongia_muta"

## [27] "Amphiprion_chrysopterus"

## [28] "Scleractinia+Sarcophyton_spp"
```

```
## [29] "Eunicella_cavolini+Eunicella_singularis+Paramuricea_clavata"

## [30] "Antipathes_dichotoma+Antipathes_grandis+Carijoa_riisei"

## [31] "Agaricia_lamarcki"

## [32] "Eunicella_singularis+Paramuricea_clavata"

## [33] "Balticina_willemoesi"

## [34] "Errina_novaezelandiae"

## [35] "Anthipatharia"

## [36] "Eunicella_cavolini"

## [37] "Montastraea_cavernosa+Siderastrea_stellata"

## [38] "Stylophora_pistillata+Symbiodiniaceae"

## [39] "Seriatopora_hystrix"

## [40] "Acropora_austera+Anthozoa"

## [41] "Ectocarpus_sp+Clodophora_sp+Lyngbya_sp+Pseudoanabaena_sp"

## [42] "Antipatharia+Octocorallia"

## [43] "Plakortis_angulospiculatus"

## [44] "+Ellisella_elongata+Octocorallia+Porifera"

## [45] "Orbicella_spp"

## [46] "Montastraea_cavernosa+Symbiodiniaceae"

## [47] "Antipatharia+Octocorallia+Porifera"

## [48] "Spinimuricea_klavereni"

## [49] "Agaricia_lamarcki+Orbicella_franksi"

## [50] "Desmophyllum_pertusum"
```

ROV & sample

```
rov <- subset(raw, grepl("rov", meso_samplingPlatform) )

rov_specimen <- subset(rov,  grepl("specimen", collectedDataType) )

print(paste("Percentage of publications relying solely on ROV where specimens
 were collected:", round(nrow(rov_specimen)/nrow(rov)*100,1),"%"))

## [1] "Percentage of publications relying solely on ROV where specimens were
 collected: 44.5 %"

rm(rov, rov_specimen)
```

Molecular techniques in taxonomic and non-taxonomic studies

```
raw$taxonomic_publication <- as.factor(raw$taxonomic_publication )

molecular <- raw %>%
  filter(molecularAnalyses != "no" &
           molecularAnalyses != "notapp" &
           molecularAnalyses != "nd")

print(paste("# Publications that adopted molecular techniques: ",nrow(molecul
ar)))

## [1] "# Publications that adopted molecular techniques:  159"

molecular_notaxa <- molecular %>%
  filter(is.na(taxonomic_publication))

print(paste("# Publications that adopted molecular techniques not for taxonom
y: ",nrow(molecular_notaxa)))

## [1] "# Publications that adopted molecular techniques not for taxonomy:  1
25"

rm(molecular, molecular_notaxa)
```

Most common target species

```
target_species <- raw %>%
  count(targetTaxa, name = "Count") %>%
  arrange(desc(Count))

target_species

## # A tibble: 440 × 2
##    targetTaxa            Count
##    <chr>                 <int>
##  1 notapp                  170
##  2 no                      112
```

```
##  3 Scleractinia             44
##  4 Corallium_rubrum         26
##  5 Porifera                 19
##  6 Corallinales             12
##  7 Paramuricea_clavata      12
##  8 Octocorallia             11
##  9 Montastraea_cavernosa    10
## 10 Rhodophyta               10
## # ℹ 430 more rows
```

```r
colnames(raw)
```

```
##  [1] "File"                          "recordTitle"
##  [3] "recordAuthors"                 "publicationYear"
##  [5] "province"                      "prov_code"
##  [7] "ecoreg"                        "scale"
##  [9] "ecoreg_code"                   "location"
## [11] "mininumSurveyedDepthInMeters" "maximumSurveyedDepthInMeters"
## [13] "latitudinalRegion"             "meso_direct_indirect"
## [15] "meso_samplingPlatform"         "taxonomic_publication"
## [17] "meso_substrate"                "meso_habitat"
## [19] "meso_protection"               "taxonomicTargetLevel"
## [21] "habit"                         "environmentalPosition"
## [23] "detected_meso_seagrass"        "detected_meso_algae"
## [25] "detected_meso_cnidaria"        "detected_meso_sponges"
## [27] "detected_meso_echino"          "detected_meso_mollusca"
## [29] "detected_meso_chondrichthyes" "detected_meso_fish"
## [31] "detected_meso_crustacea"       "detected_meso_other_taxa"
## [33] "indicators"                    "collectedDataType"
## [35] "meso_protocol_campaign"        "samplingUnit"
## [37] "visuallySurveyedSurface"       "meso_design"
## [39] "meso_replicates"               "temporalReplicates"
## [41] "abioticVariablesCollection"    "abioticVariables"
## [43] "molecularAnalyses"             "targetTaxa"
## [45] "DOI"                           "meso_dimensions"
## [47] "observationalORexperimental"   "max_depth_direct"
## [49] "max_depth_hov"                 "max_depth_rov"
## [51] "min_depth_hov"                 "min_depth_rov"
## [53] "max_depth_ccr"                 "max_depth_open"
## [55] "min_depth_ccr"                 "min_depth_open"
## [57] "max_depth_scuba"               "min_depth_scuba"
## [59] "field_cons"                    "field_commdesc"
## [61] "field_map"                     "field_impass"
## [63] "field_troph"                   "field_taxo"
## [65] "field_dist"                    "field_aqua"
## [67] "field_method"                  "field_physio"
## [69] "field_func"                    "field_habtype"
## [71] "field_fishery"                 "field_stock"
## [73] "field_paleo"                   "field_phylo"
## [75] "field_rest"                    "field_archaeo"
```

```
## [77] "field_micro"                    "field_biomedicine"
## [79] "field_chem"                     "field_Molecular_ecology"
## [81] "field_Acoustics"                "field_environmental_drivers"
## [83] "field_Behaviour"                "field_Geomorphology"
## [85] "field_Taxonomy"
```

```
print(paste("Publications on Corallium rubrum :", nrow(subset(raw, grepl("rub
rum", targetTaxa)))))
```

```
## [1] "Publications on Corallium rubrum : 32"
```

```
print(paste("Publications on Paramuricea clavata:", nrow(subset(raw, grepl("c
lavata", targetTaxa)))))
```

```
## [1] "Publications on Paramuricea clavata: 27"
```

```
print(paste("Publications on Montastraea cavernosa:", nrow(subset(raw, grepl
("cavernosa", targetTaxa)))))
```

```
## [1] "Publications on Montastraea cavernosa: 27"
```